

1.3.1. COG/SOG/HDG/STW Window

HDG	270.0°	STW	5.0 kn	COG	270.0°	SOG	5.0 kn
GYRO 1		LOG 1		GPS 1		GPS 1	

COG/SOG/HDG/STW window is designed for the display of ship motion parameters. The panel displays the following parameters:

- **HDG** – course from gyrocompass;
- **STW** – ship speed through the water;

NOTE: Only longitudinal part if DLOG is selected.

- **COG** – ship course over the ground;
 - **SOG** – ship speed over the ground.
-

NOTE: Calculated part (longitudinal and transversal) if DLOG is selected.

The data source is specified to the right of the own ship motion parameter name:

- **GYRO1** – gyrocompass with a digital input, its number specified;
- **MAG** – magnetic compass with a digital input;
- **GPS1** – GPS (DGPS), its number specified;
- **ER** – echo reference mode;
- **DR** – dead reckoning mode;
- **MAN** – manual input;
- **LOG1** – log (Speed Through Water) with a digital output, its number specified;
- **DLOG1** – Doppler log, its number specified;
- **NONE** – no source available.

HDG	270.0°	STW	5.0 kn	COG	270.0°	SOG	5.0 kn
GYRO 1		LOG 1		GPS 1		GPS 1	

VEGA

As a free cursor is positioned on the source name, the source alias entered in the System Configuration utility is displayed on the pop-up prompt window.

HDG	292.5°	STW	12.0 kn	COG	292.2°	SOG	12.0 kn
GYRO 1		DLOG 1		GPS 1		GPS 1	

Gyro offset: +0.3°

If there is an offset to the gyro, the window displays corrected heading and a special symbol: the yellow triangle. As a free cursor is positioned on this triangle, the correction value is displayed on the pop-up prompt window.

For the data display principles, see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 2, section Navigational Sensors, paragraph Display of Navigational (Essential) Information.*

1.3.2. Alerts Window



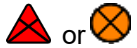
The **Alerts** window is intended for displaying the alarms, warning and cautions. The window contains the following units:

-  – to open a list of generated alarms;
-  – to open a list of generated warnings;
-  – to open a list of generated cautions.

Given below is an example of a caution:



Category A alerts from the RADAR source can be acknowledged in the RADAR task (Navi-Radar 4000 application) only. Category A alerts from the ECDIS source can be acknowledged in the ECDIS task (Navi-Sailor 4000 application) only. Even if it is not permitted to acknowledge an alarm in the ECDIS or RADAR task, it will nonetheless be displayed in the task's alert list. An alert which cannot be acknowledged is shown with the

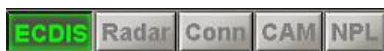


or icons ("Can't be acknowledged"):



For more detailed description of the alerts display, see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Display of Alerts and section Alert Management Principles, paragraph General*.

1.3.3. Application Window



The **Applications** window is intended for loading/switching MFD applications:

- **ECDIS** – to turn on/switch to the ECDIS task;
- **Radar** – to turn on/switch to the RADAR task;
- **CONN** – to turn on/switch to the Conning Information Display task;
- **CAM** – to turn on/switch to the Central Alert Management (Previous BAM) task;
- **NPL** – to turn on/switch to the Navi-Planner 4000.

1.3.4. Centered Mode Window



The window contains the following controls:

- **To Centered Mode** – button with the message meaning that the ship symbol is not displayed on the chart. With a press on the button, the display of the ship symbols is restored. After the restoration of the ship symbol display (also with the use of **Ahead** function), Wärtsilä logo appears in the place of a message.

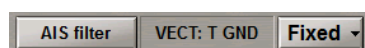
1.3.5. Sensors Window



The **Sensors** window is intended for the display of information from different sources. The window contains the following controls:

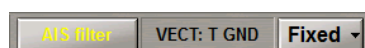
- **X1** – to select a scanner which the radar picture is supplied from;
- **Radar ovr** – to turn on the radar overlay information display on the electronic chart in the ECDIS;
- **AIS** – to turn on the display of AIS targets;
- **ARPA** – to turn on the display of information on targets processed by the ARPA and to select ARPA target sources.

1.3.6. Vectors Window



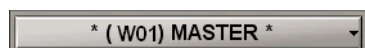
The **Vectors** window is designed for the display of the following parameters:

- **AIS filter** – to promptly open the **AIS** page of the **Targets** display on the Control panel. If the sleeping AIS targets filtering is enabled, or the display of some AIS target type is turned off (the filter is on this page), the button name is shown in yellow (see also document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Indications, paragraph Display of Important Indications*);



- **VECT: T GND** – indication showing the AIS targets vector mode and stabilization mode (see also document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Indications, paragraph Display of Important Indications*);
- **Vectors** – to call a menu for setting lengths of the own ship and target motion vectors; the **Fixed** option allows a constant vector length to be set (in proportion to the screen size), the rest – by the set time value (in minutes, 1– 24 min).

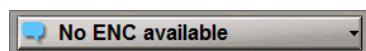
1.3.7. Network Window



The **Network** window is intended for displaying the station name and status in the network and, in the case of the MASTER status, also the transfer of rights to other stations in the network. The window contains the following control:

- **(W01) MASTER** – to display the station name and status in the network.

1.3.8. Indications Window

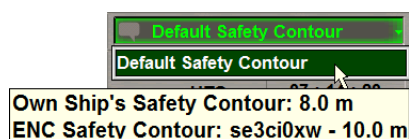


The **Indications** window is designed for the display of indications.

As a new indication appears, the drop-down list button displays the  sign. Press the button for familiarizing yourself with all the indications.

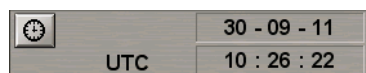


As a free cursor is positioned on the **Default Safety Contour** indication, more detailed explanation for this indication is displayed in the pop-up prompt window.

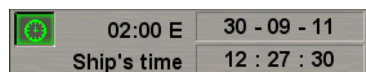


For more detailed description of the indications display, see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Indications, paragraph Display of Indications.*

1.3.9. Time Window



The **Time** window is intended for the display of the current time. There may be two types of time values in the window: UTC and ship's time. To switch between them, press the button with a picture of the clock face (ship time if the button is pressed, UTC if it is released):



The window contains the following elements:

- **UTC/Ship's** – information on which time is currently displayed;
- **00:00 W/E** – time zone or the difference between the ship time and UTC (not displayed if UTC is selected); the format is hh:mm;
- **01-01-01** – display of the current date according to the selected time type; the format is dd-mm-yy;
- **00:00:00** – display of the current time; the format is hh:mm:ss.

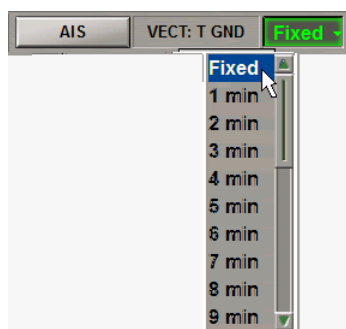


Select from the drop-down lists the following settings:

- **Ship by ...** – to display the own ship (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 3, section Ship Motion Data, paragraph Ship Symbol on the Electronic Charts*) on the Chart panel as:
 - **symbol** – the ship symbol (regardless of the scale);
 - **contour** – the ship contour (when the current screen scale is commensurate with the ship dimensions).
- **Align by ...** – to switch orientation (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 3, section Ship Motion Data, paragraph Own Ship Motion Vectors*) of the own ship symbol display along:
 - **HDG** – the vector of the ship's motion as per the heading detector (gyro) reading;
 - **COG** – the ship's true motion (COG) vector.
- **Wind vector** – to select true wind vector display mode (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 2, section Other Sensors, paragraph Wind Interface*).

3.6.6. Setting of Lengths of Own Ship and Target Vectors

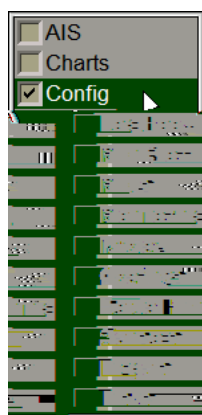
Press vectors button in the **Vectors** window on the Control panel (the button shows the current vector length).



Select the required vector length (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 3, section Own Ship Motion Data, paragraph Own Ship Motion Vectors*) from the list which will open up and press the left trackball button.

3.7. Setting Time Zone

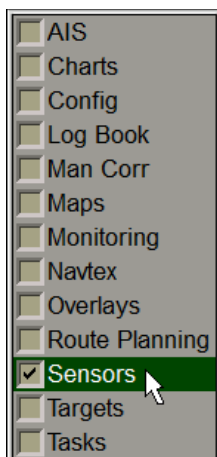
Open the **Config** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



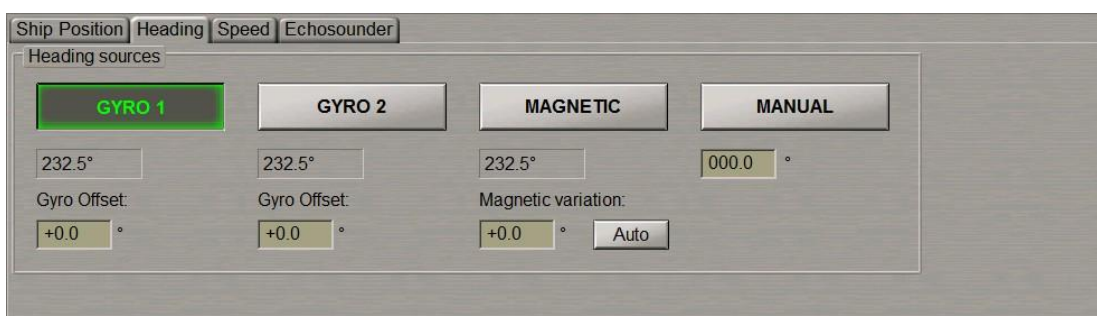
4.2. Setting Heading Source

ATTENTION! Setting of heading sources is available only at the station with the MASTER status.

Open the **Sensors** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Sensors** panel to switch to the **Heading** page:



The **Heading** page is designed for selection of a heading source. The **Heading** page shows all the connected heading sensors.

The group of each heading source contains the following elements:

- **GYRO 1** – to select this sensor as the heading source; the window is intended for displaying a heading value from the sensor;
- **Gyro Offset** – for the manual input of the gyro offset (for gyro compass);
- **Magnetic variation** – for the input of the magnetic variation from the database (the Auto button), or manually in the input window (for the magnetic compass);
- **MANUAL** – to select the use of manual heading input; the window is intended for the heading input.

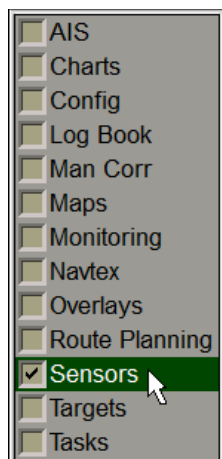
Press the button with the source name to select the main source of heading whose data will henceforth be used for making all the calculations in the system.

If necessary, enter errors (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 2, section Navigational Sensors, paragraph Compensations for Navigational Data*) for gyro and variation for magnetic compass.

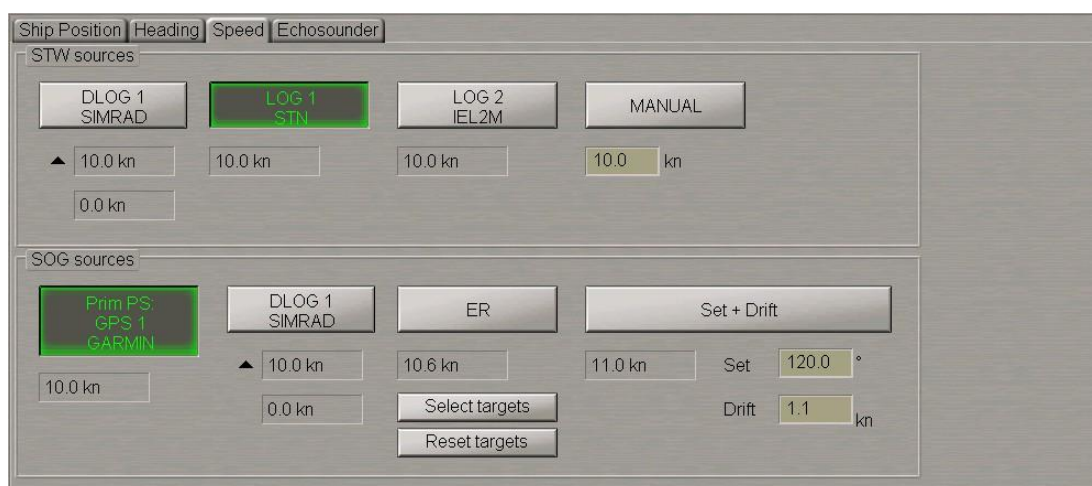
4.3. Setting Speed Sources

ATTENTION! Setting of speed sources is available only at the station with the MASTER status.

Open the **Sensors** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Sensors** panel to switch to the **Speed** page.



The **Speed** page is designed for selecting a speed source.

4.3.1. Setting Speed-Through-Water Source



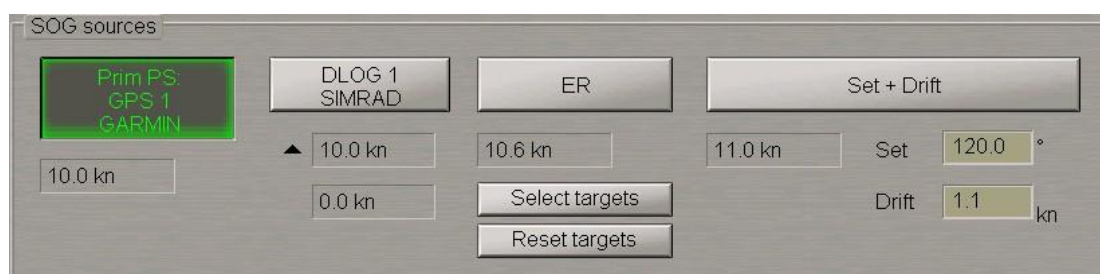
The **STW sources** group is designed for selecting a source of speed through the water. The group shows all the connected speed-through-water sensors.

The group of each speed-through-water source can contain the following elements:

- **DLOG** – to select the DLOG as a source of STW; the boxes are intended for displaying longitudinal and transverse components of the speed-through-water vector, the arrows showing the direction of components;
- **LOG** – to select the LOG as a source of speed through water; the window is intended for displaying STW value from the log;
- **MANUAL** – to select the use of manual STW input; the window is intended for the speed input.

Press the button with the source name to select the source of speed through the water, whose data will henceforth be used for making all the calculations in the system.

4.3.2. Setting Speed-Over-Ground Source



The **SOG sources** group is designed for selecting a source of speed over ground. The group shows all the connected speed-over-ground sensors.

The group of each speed-over-ground source can contain the following elements:

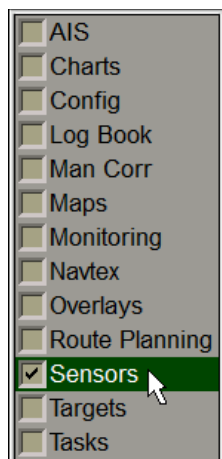
- **Prim PS: GPS 1** – to select the EPFS as a source of SOG; the window is intended for displaying SOG value from the EPFS;
- **DLOG** – to select the DLOG as a source of SOG; the boxes are intended for displaying longitudinal and transverse components of the SOG vector, the arrows showing the direction of components;
- **ER** – to select the ER mode as a source of SOG; the window is intended for displaying the SOG value; the buttons are intended for:
 - **Select targets** – to select targets as fixed reference points (the button is enabled if at least one steadily tracked target is available);
 - **Reset targets** – to cancel reference points.
- **STW + Drift** – to select STW as SOG, manual drift input taken into account. The window is intended for displaying SOG value. The input boxes are intended for:
 - **Set** – to display and manually enter the set;
 - **Drift** – to display and manually enter the drift.

Press the button with the source name to select the source of speed over ground whose data will henceforth be used for making all the calculations in the system.

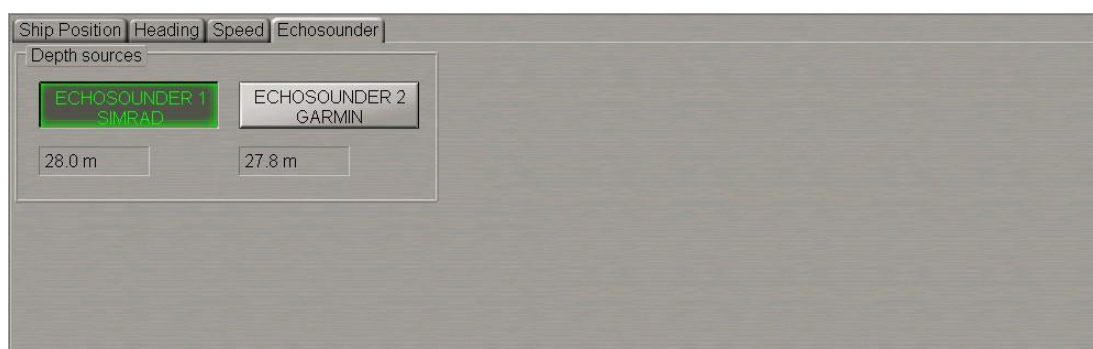
4.4. Setting Depth Source

ATTENTION! Setting of depth sources is available only at the station with the MASTER status.

Open the **Sensors** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Sensors** panel to switch to the **Echosounder** page.



The **Echosounder** page is designed for selecting a depth source. The **Echosounder** page shows all the connected depth sensors.

The group of each depth source can contain the following elements:

- **ECHOSOUNDER 1** – to select this sensor as a depth source; the window is intended for displaying the depth value from the sensor.

Press the button with the source name to select the main source of depth whose data will henceforth be used for making all the calculations in the system.

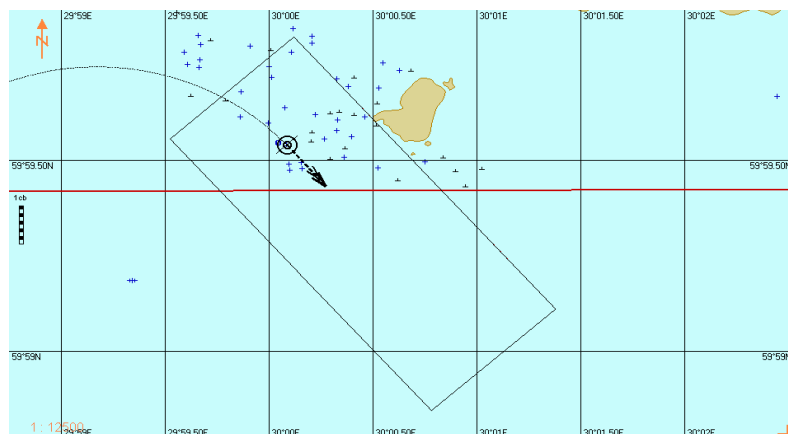
5.2.1. Setting of Safety Parameters for Detection of Dangerous Objects

The **Look-Ahead frame** group is intended for setting the size of the frame, which will be used for the chart data analysis and for the generation of the Antigrinding alerts, Area alerts and Navigational alerts:

- **Ahead** – window for the input of advance time for alert generation. The time value determines the length equal to the distance covered by the ship proceeding at the current SOG (from 30 s to 60 min);
- **Port** – to set the width of the corridor to the left of the ship (from 0.1 to 4.0 NM);
- **Starboard** – to set the width of the corridor to the right of the ship (from 0.1 to 4.0 NM);
- **Show Look-Ahead frame** – to turn on the display of a safety frame on the ECDIS task screen.

In the **Ahead**, **Port** and **Starboard** input boxes, set the necessary values which will determine the shape of the safety frame.

Select the **Show Look-Ahead frame** checkbox to show the safety frame. The safety frame is displayed with a black outlined rectangle:



The **Safety parameters** group is intended for setting and viewing safety parameters in the Navigation Mode:

- **Shallow contour** - in the ENC, the four shades mode defines the value of the shallow contour affecting colours of depth areas (dark blue for depth areas with values shallower than the shallow contour);
- **Safety contour** - mariner-selected contour to distinguish on the display between safe and unsafe water;
- **Safety depth** - mariner-defined depth to emphasize soundings on the display equal to or less than this value;
- **Deep contour** - in the ENC, the four shades mode defines the value of deep contour affecting colours of depth areas (the lightest S-52 colour for depth areas deeper than the deep contour).

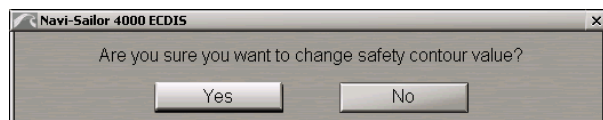
Use the **Shallow contour** line to enter the value of the shallow contour affecting colours of depth areas for ENC format charts:

Shallow contour: m

Use the **Safety contour** line to enter the safety contour value:

Safety contour: m

Press **Yes** to confirm the changes you have made:

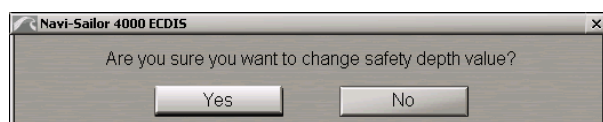


NOTE: The alert is generated exclusively by the safety contour value, even if the shallow contour value is smaller than the safety contour value.

Use the **Safety depth** line to enter the safety depth value:

Safety depth: m

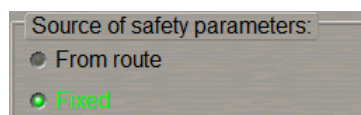
Press **Yes** to confirm the changes you have made:



Use the **Deep contour** line to enter the deep contour value affecting colours of depth areas for ENC format charts.

Deep contour: m

In the **Source of safety parameters** group, select the source which the safety parameters will be taken from for the route monitoring:



If the route is not loaded for monitoring, the **Fixed** radio button will be selected automatically, safety parameters set in the **Safety parameters** group are used for monitoring dangers.

As a route to be monitored is loaded, the last operator-selected option ('**From route**' or '**Fixed value**') is set automatically.

If the **Fixed** value is selected, the safety parameters set in the **Safety parameters** group are used (see also document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Main Alerts Generation Principles, paragraph Safety Alerts*).

If the **From route** value is selected, safety parameters are selected by using the procedure detailed in *Chapter 4, section Loading of Route and Schedule in the Navigation Mode, paragraph Selection of Safety Parameters when Loading Monitored Route*).

5.2.2. Setting of Safety Parameters in Sailing in the Vicinity of Dangerous Objects

NOTE: The **Antigrounding alerts** group contains checkboxes for enabling/disabling the following safety parameters excluded from the version 3.02.350.9047 and alerts by default in ON state

BEFORE	AFTER R10 3.02.350.9047
	

- **Safety contour** – to enable the alarm generation upon the approach to the depth line less than the **Safety contour** value;
- **Nav. hazard** – to enable the caution/warning generation upon the approach to the navigational hazards (for the list of objects see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Main Alerts Generation Principles, paragraph Safety Alerts*);
- **Highlight Danger** - to turn on the highlight the dangers on the Chart panel.

Select the **Safety contour** checkbox to turn on the alarm generation as the safety contour is crossed by the safety frame.

Select the activated **Nav. hazard** checkbox to enable the caution (warning) generation when a navigational hazard turns up within the safety frame (see also paragraph *Setting Priority for Navigational and Area Alerts*).

Select the **Highlight Danger** checkbox to turn on the highlighting of dangers on the Chart panel falling within the safety frame.

NOTE: If at least one checkbox in the **Antigrounding alerts** group is cleared, the Chart panel displays the **SF CNT** important indication (for details see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Indications, paragraph Important Indications*).

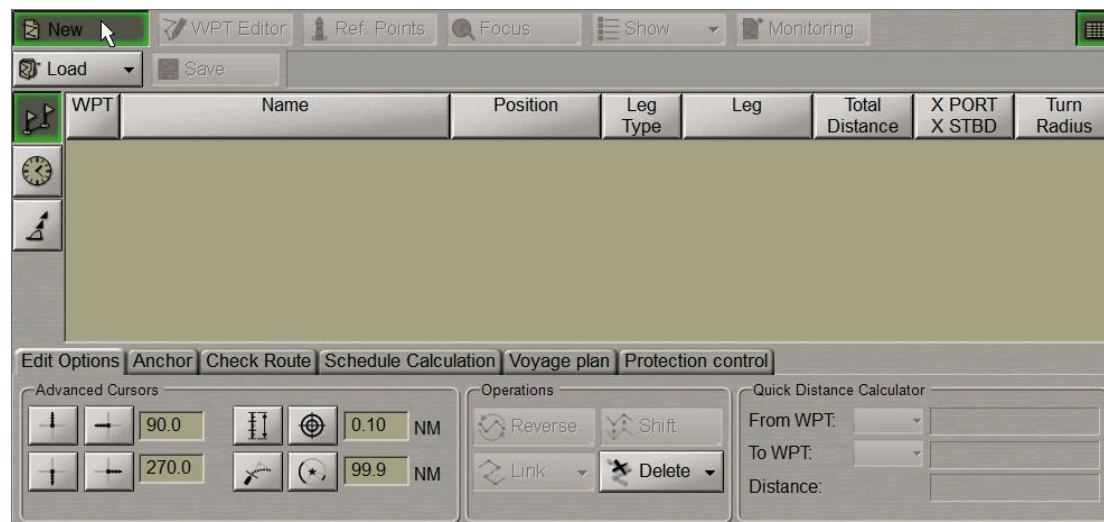
5.2.3. Monitoring of Crossing of Area Limits in IMO and Additional Groups

The **IMO Areas** tab containing checkboxes for the selection (assigning danger-to-navigation status) from the main areas tracked in accordance with IEC-61174 standard.



9.1.2.2. Plotting Route

Press the **New** button in the top left part of the panel to display a blank table for the generation of a new route.



The graphics cursor will appear on the Chart Panel, whereas the right bottom part of the screen will display the **Place First Point** information window.

Place first point

Latitude58°09.781' N
Longitude011°53.543' W
Anc. Brg000.0 °
Anc. Rng0.00 NM

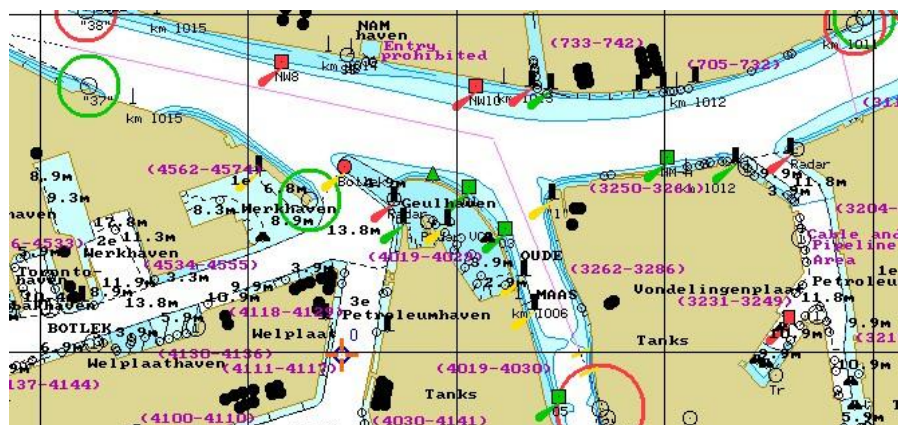
Ship position

true
relative

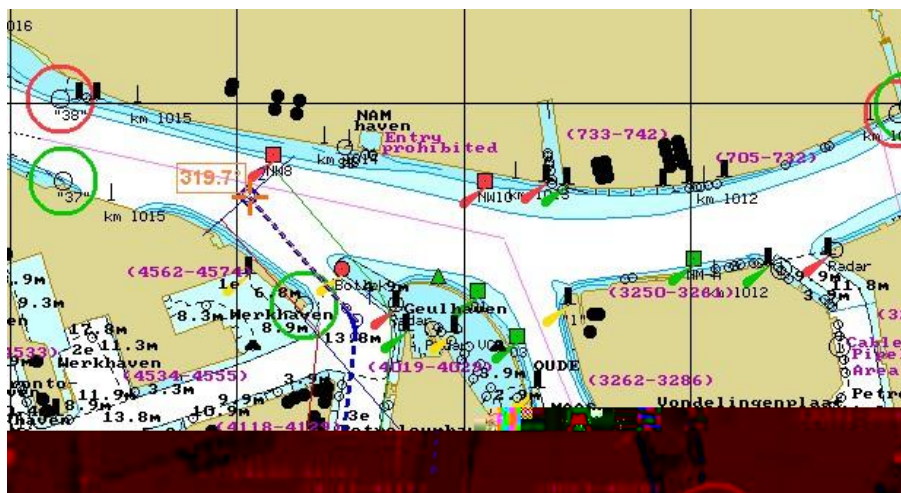
Bearing272.3020.8 °
Opp. Bearing092.3159.2 °
Range757.85 NM

Press Tab to edit manually

Set the cursor in the start point coordinates by moving the trackball, using the data in the **Place First Point** window, and press the left trackball button.

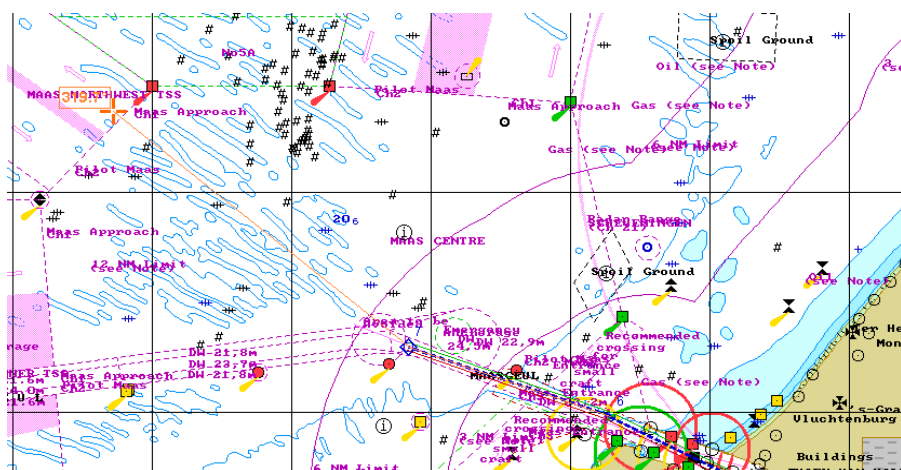


The symbol of the start point with its number ("0") will appear on the Chart Panel. Position the cursor in the coordinates of the next point. At this stage, a route leg line will be drawn between the cursor and the start point. Press the left trackball button: the Chart Panel will display the next point symbol with "1" for number and the plotted route leg. By default, XTD lines are also shown. To turn off their display, use the **Display** button (see below).

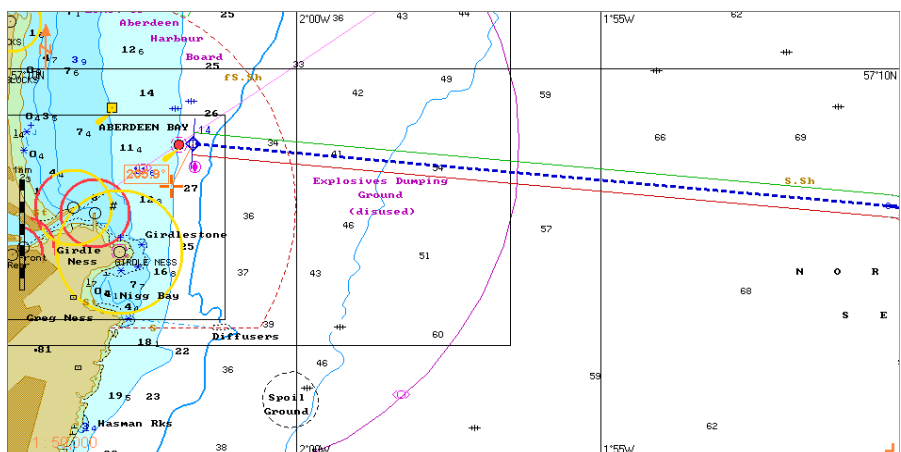


If it is necessary to set several waypoints, repeat the trackball moving and left button pressing operation as many times as there are WPT's required to be set.

If the cursor is positioned beyond the Chart Panel boundaries, it will be re-drawn automatically so that there is always a chart from the chart folio under the cursor.



After the setting of the last point of the planned route, double click the right trackball button: the cursor will exit from the route planning and editing mode and will assume the form of the free cursor.



The generation of the route is completed.

9.1.2.3.Entering Additional Data

The display of the table is turned on/off by using the **Show tables** button.



In the left part of the **Route Planning** panel, press the button to display the corresponding table.



The screenshot shows the 'Route Planning' panel with the 'Waypoints' table displayed. The table has columns: WPT, Name, Position, Leg Type, Leg, Total Distance, X PORT, X STBD, and Turn Radius. The table contains four rows of data. Below the table are several control panels: 'Edit Options' (Anchor, Check Route, Schedule Calculation, Voyage plan, Protection control), 'Advanced Cursors' (with directional buttons and numerical inputs), 'Operations' (with buttons for Reverse, Shift, Link, and Delete), and a 'Quick Distance Calculator' (with inputs for From WPT, To WPT, and Distance).

WPT	Name	Position	Leg Type	Leg	Total Distance	X PORT	X STBD	Turn Radius
0		51° 53.659 N 004° 18.420 E	XX	XXX.X XXX.X	XXX.X	XXX.X	XXX.X	XXX.X
1		51° 54.865 N 004° 14.838 E	RL	298.6° 2.27 NM	2.27 NM	0.10 NM	0.10 NM	0.50 NM
2		51° 56.571 N 004° 11.176 E	RL	307.0° 2.84 NM	5.11 NM	0.10 NM	0.10 NM	0.50 NM
3		51° 59.480 N 004° 02.548 E	RL	298.6° 6.09 NM	11.21 NM	0.10 NM	0.10 NM	0.50 NM

The **Waypoints** table includes the following data (by columns):

- **WPT** – WPT number;
- **Name** – WPT name;
- **Position** – WPT latitude and longitude coordinate;
- **Leg Type** – route leg type (RL or GC) from the previous to the current WPT set by the operator; all route leg calculations are referenced to the WGS-84 reference datum;
- **Leg** – course and distance from the previous to the current WPT;
- **Total Distance** – summary distance between the initial (0) and current WPT by the route legs;
- **X PORT/X STBD** – safe zone to the left and to the right of the route leg;
- **Turn Radius** – operator-set radius of the arc for the display of an approximate ship track on the turn.

For input of the additional data, perform the procedures detailed below:

- position the cursor on the **Name** element cell; double click the left trackball button to activate the cell;
- enter the WPT name; press <ENTER>;
- position the cursor on the **Leg Type** element cell; double click the left trackball button to activate the cell;
- press the button in the right-hand part of the cell and select the sailing type from the list which will open up;
- press the left trackball button to confirm the changes you have made;
- position the cursor on the **XTL PORT/XTL STBD** element cell; double click the left trackball button to activate the cell;
- edit the XTD value to the left of the route leg; press the left trackball button;
- edit the XTD value to the right of the route leg; press the left trackball button;
- position the cursor on the **Turn Radius** element cell; double click the left trackball button to activate the cell;
- enter the radius for the arc of the suitable ship track on the turning circle; press the left trackball button.

Name

Name
St-Petersburg

Leg type
RL

Leg type
RL
RL
GC

Leg type
GC

X PORT	
X STBD	
0.10	
0.10 NM	

X PORT
X STBD
0.12 NM
0.12

X PORT
X STBD
0.12 NM
0.12 NM

Turn radius
0.10

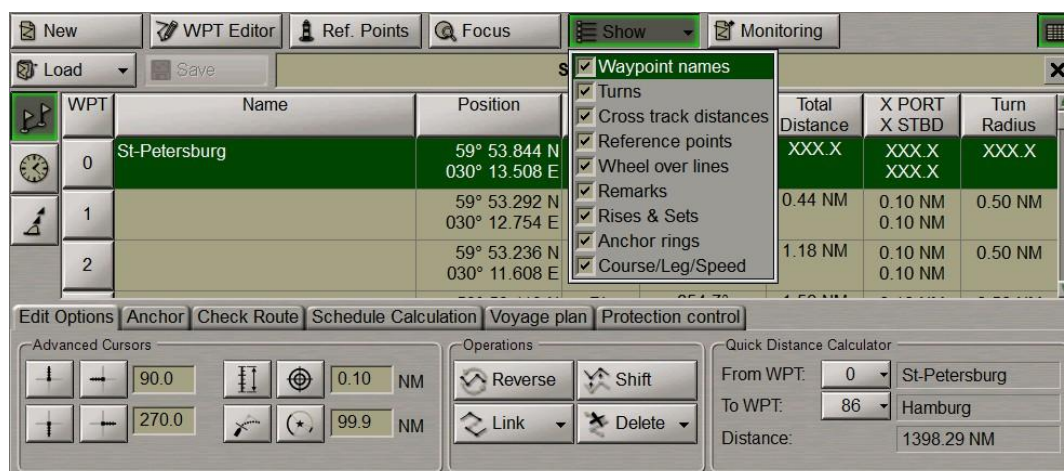
Turn radius
1.00 NM

When filling in the table, you can use the context menu called by the right trackball button:

- **Clear cell** – to delete a value in the selected cell;
- **Fill cells down** – to fill in all the cells below with a value entered in the selected cell;
- **Fill cells up** – to fill in all the cells above with a value entered in the selected cell.

Total Time	Speed
	1
	100%
	Copy waypoint
	Out waypoints
	Delete waypoint
	Insert waypoint
	Clear cell
	Fill cells down
	Fill cells up

Press the **Show** button in the top part of the panel.



The following checkboxes in the drop-down list for turning on the display of route options on the ECDIS task screen are selected by default:

- **Waypoints names** – to display on the electronic chart WPT names;
- **Turns** – to display on the electronic chart the turn radius arc for the active route;
- **Cross track distances** – to display on the electronic chart safety lines for the active route;
- **Reference points** – to display Reference points;
- **Wheel over lines** – to display on the electronic chart WOL marks;
- **Remarks** – to display remarks;
- **Rises & Sets** - to display sun and moon rise and set symbols;
- **Anchor rings** - to display anchor rings;
- **CRS/DIST/SPD** - to display course, distance and speed for each route leg.

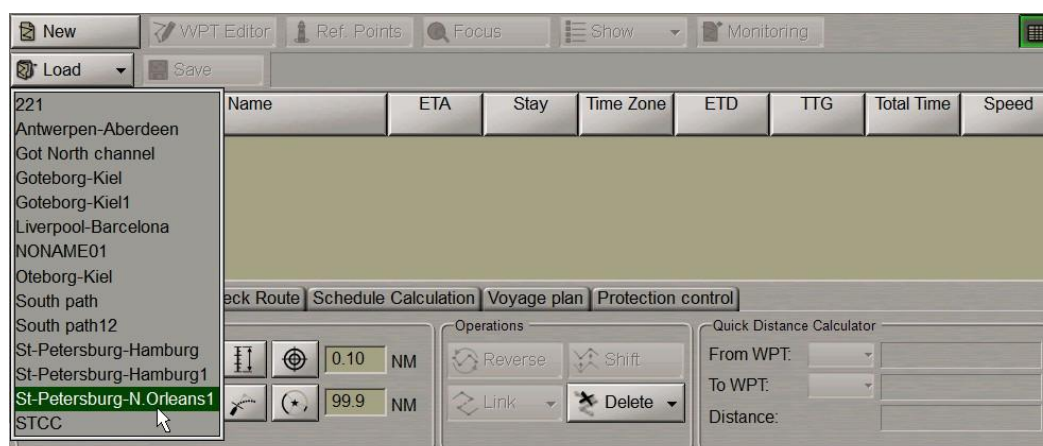
If necessary, clear the **Waypoints names** checkbox to turn off the display of waypoints' names on the ECDIS task screen.

To save the route, enter its name in the name input field and press the **Save** button.

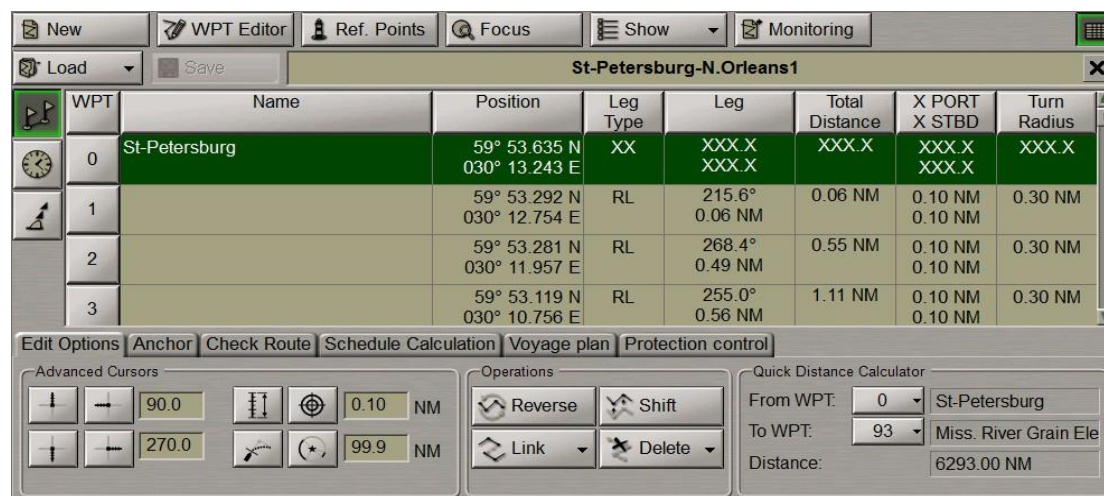


9.1.3.Route Loading and Unloading

In the top left part of the **Route Planning** panel, press the **Load** button.



In the list which will open up select the route required to be loaded in the editing mode and press the left trackball button.



Several routes can be loaded simultaneously in the editing mode. As a button with route name is pressed, the route becomes active and can be edited on the screen by using the graphic editor.



To quick load the necessary route in the monitoring mode, select it from the routes loaded in the editing mode and press the **Monitoring** button (see also *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). User Manual document, Chapter 4, Loading of Route and Schedule in the Navigation Mode* section and section *What is Different in Route Monitoring in the Route Planning* of this document):



To unload the route, press the  button to the right of the route name.

9.1.4. Fast Route Search

Load the necessary route (see item *Route Loading and Unloading*).

In the top part of the **Route Planning** panel, press the **Focus** button.

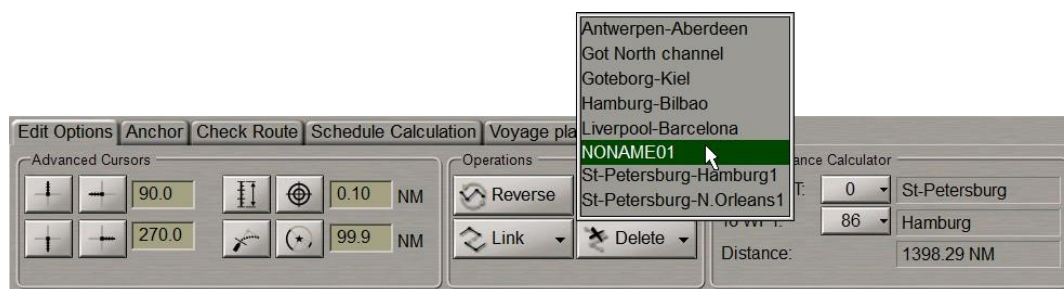


The Chart Panel will display the entire route on the suitable scale chart.

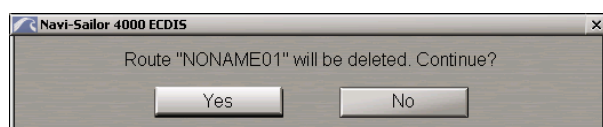


9.1.7. Deleting Route

Switch to the **Edit Options** page. Press the **Delete** button in the **Operations** group.



Select the route which should be deleted from the list and press the left trackball button.



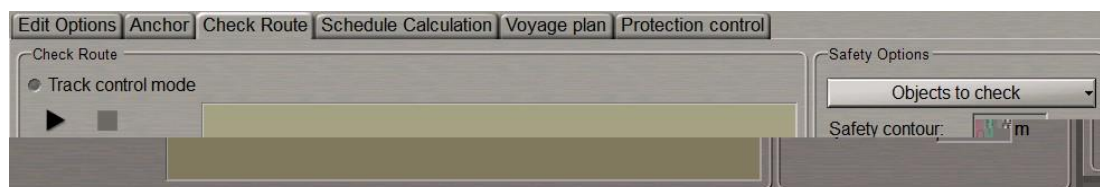
In the window which will appear, press Yes to confirm the deleting. The route will be deleted.

ATTENTION! The route cannot be deleted until it is loaded into monitoring or editing mode.

9.1.8. Checking and Editing Route Taking into Account Dangers to Navigation

Load the route which should be checked for the availability of dangers to navigation (see *Route Loading and Unloading* item).

Switch to the **Check route** page.

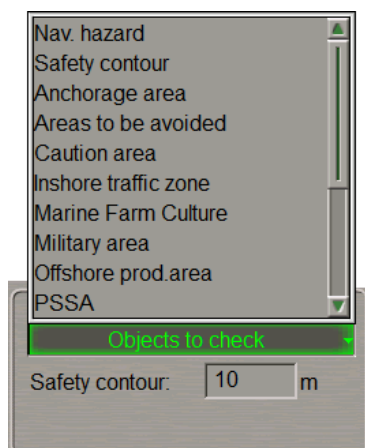


In the **Safety options** group select which charts should be used for the check only charts with the **Best scale** or all the charts (**All scales**) under the checked route (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Main Alerts Generation Principles, paragraph Safety Alerts*).

Press the  button in the **Check route** group: the route will be checked.



The list of chart objects and safety contour values used in the route check are displayed in the **Safety options** group.



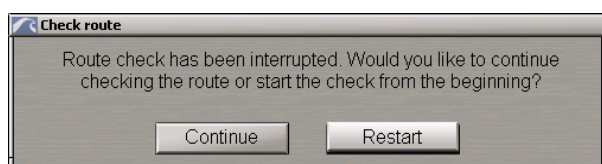
The check process is shown in the display window.



Press the  button if it is necessary to interrupt manually the check of the entire active route for the availability of dangers to navigation.

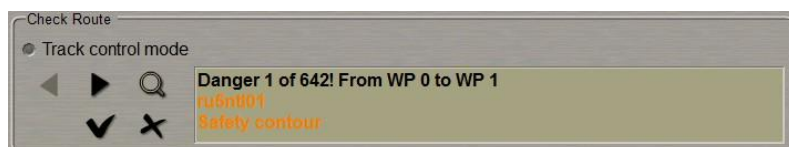


The route check will be interrupted. It can be resumed after the necessary time without exiting from the route checking mode. To do this, press the  button.

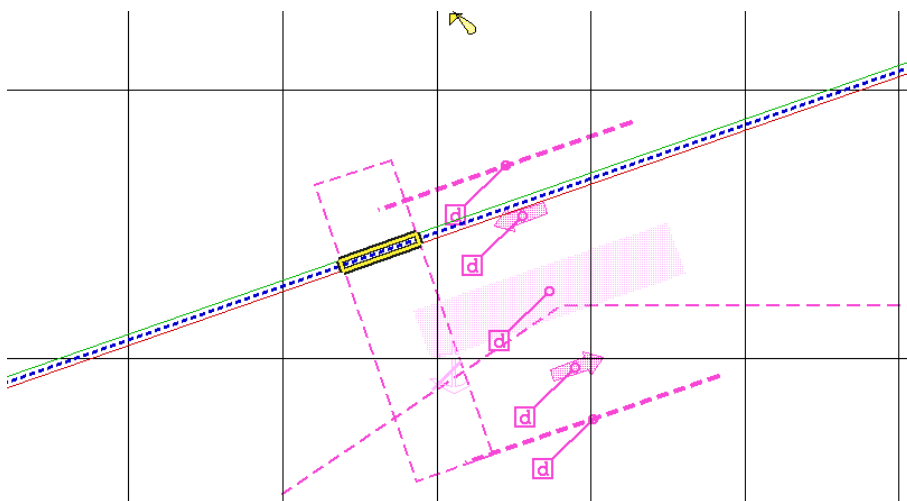


Select the necessary option, the route check will be resumed.

After the end of check, the **Check route** group will display a message about detected dangers or about no dangers having been found.

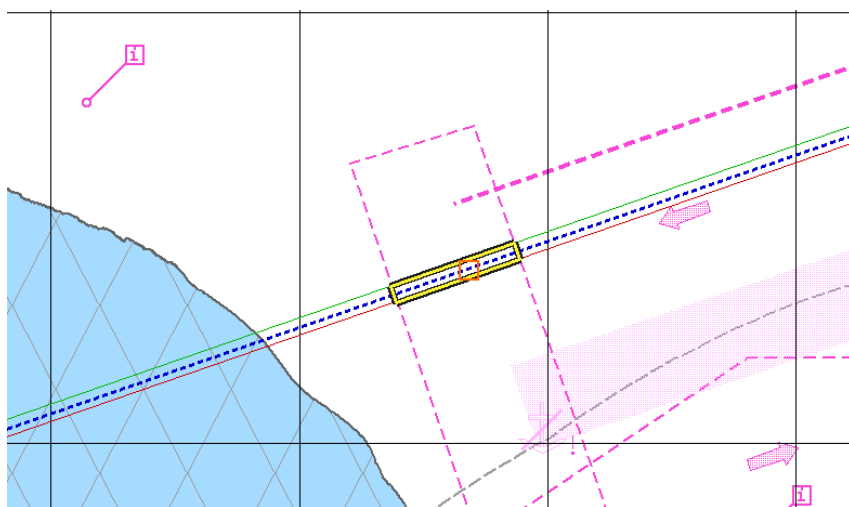


To view the detected dangers to navigation, use the  and  buttons to select the danger you are interested in. Press the  button. The danger to navigation will be indicated by highlighting on the Chart Panel. In this case the best chart on the original scale will be loaded.

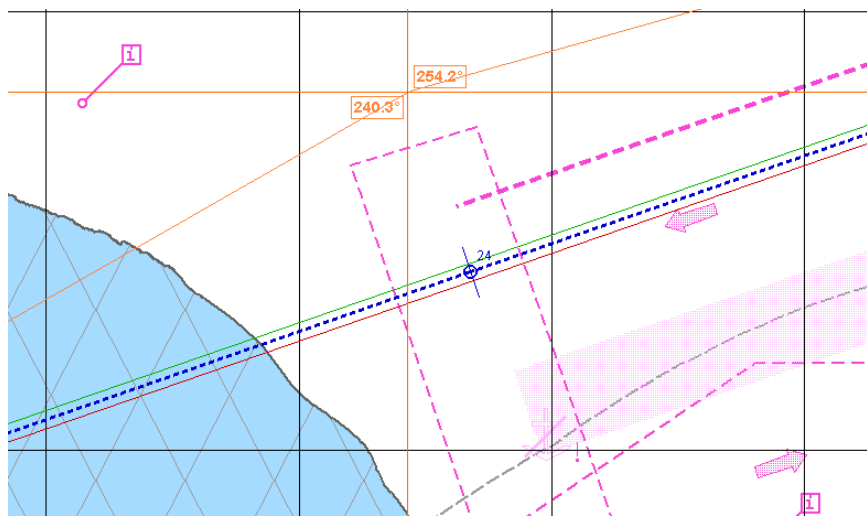


To edit the route to take into account the detected dangers to navigation, use the  and  buttons to select the required danger and press the **WPT Editor** button.

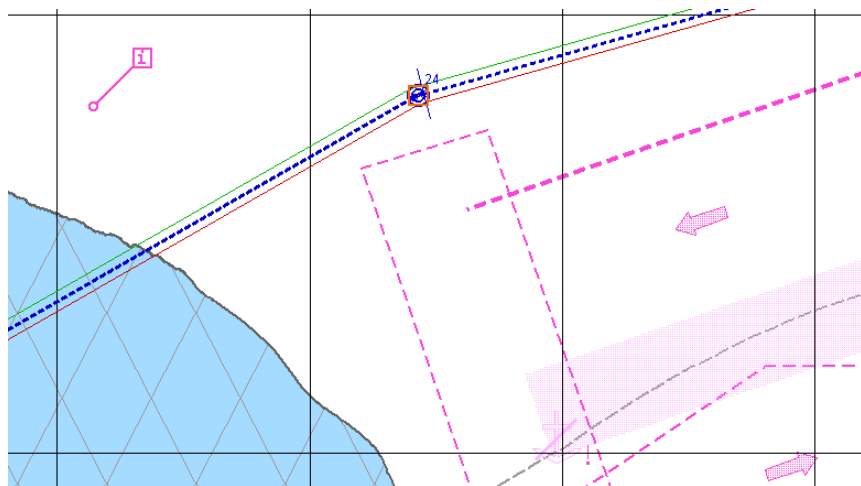
An acquisition marker will appear on the Chart panel in the point with coordinates of the detected danger to navigation.



Edit the route leg according to the editing rules so that the route leg does not pass through the dangerous place. E.g., change the WPT position.



After the route leg editing is completed, press the left trackball button. After the change of the WPT position, two legs before the changed point and two legs after it will be automatically checked.

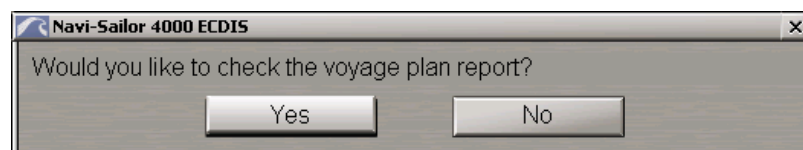


Perform the editing until the end of the route (see also section *Route Editor*, paragraph *Graphic Route Editing Method*).

If there are no dangers to navigation on the route, the **Check route** group will display an appropriate message.



To accept the route checking results and exit from the checking mode press the  button. The "checked" attribute will be assigned to the route. Check results will be entered in the voyage plan, the following window will open up.



To view check results, press the **Yes** button.

Print Preview

PDF HTML Excel

Page 1 of 35

Voyage No.: Ship's Name: KOLA FROM: TO: Master's Name:

WAYPOINTS									
WPT	Name	Position	Leg	Total Distance	X PORT X STBD	Turn Radius	Draught	Squat	UKC
0	St-Petersburg	59° 53.844 N 030° 13.509 E	XXXX X XXXX X	XXXX X	XXXX X XXXX X	XXXX X XXXX X	XXXX X XXXX X	XXXX X XXXX X	XXXX X
1		59° 53.292 N 030° 12.754 E	214.5° 0.22 NM	0.22 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
2		59° 53.236 N 030° 11.608 E	264.4° 0.74 NM	0.96 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
3		59° 53.119 N 030° 10.756 E	254.7° 0.32 NM	1.28 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
4		59° 56.852 N 029° 52.510 E	282.2° 10.07 NM	11.35 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
5	Pilot buoy	59° 57.756 N 029° 47.762 E	280.8° 2.46 NM	13.81 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
6		59° 58.845 N 029° 45.408 E	312.7° 1.59 NM	15.40 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
7		59° 59.094 N 029° 43.732 E	286.5° 0.96 NM	16.36 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
8		59° 59.309 N 029° 40.950 E	278.8° 1.41 NM	17.77 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
9		60° 01.736 N 029° 25.785 E	287.7° 7.97 NM	25.74 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	Shallow water
10		60° 03.242 N 028° 31.410 E	273.2° 27.25 NM	52.99 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	9.5 m
11		59° 59.961 N 028° 52.897 E	273.9° 5.00 NM	107.72 NM	0.10 NM 0.10 NM	0.50 NM 0.10 NM	10.0 m 0.5 m	0.5 m	9.5 m

Printed: 12/12/15 17:35:17 Prepared by nav. Officer: Checked by Master: Page 1

The Voyage Plan table on the **Safety Parameters** page specifies safety parameters which were used for checking each route leg. The **List of Accepted Alerts** column lists dangers which were found during the route check but were ignored by the user. If no dangers were found on these legs, cells of the **List of Accepted Alerts** column will be empty. Specified in the brackets are charts on which dangers were detected, as well as danger depths for the safety contour and safety depth. The Voyage Plan footer displays: **Check for safety: 07.10.12 13:25 (date and time of the check)**.

To save check results, export the voyage plan in one of the formats (PDF HTML Excel) by pressing the respective button in the left top corner (PDF, HTML, Excel) of the **Print Preview** window.

ATTENTION! As the route is unloaded, data in the **List of Accepted Alerts** column is not saved.

The saved files have the `*.hba` format specifying the status (checked), check time and date.

To exit from the checking mode without accepting the route check results, press the  button.

For checking route geometry for compliance with the Track Control Mode, press the **Track control mode** radio button.

Check route

☐ ☒ Track control mode

For the reference purpose, the **Track control** group specifies limitations for the route elements required for the Track Control mode operation.

Track control

WOL distance: 130 m

Min. turn radius: 0.50 NM

Max. XTD: 1.00 NM

Use the same procedure to check the route as described above for safety parameters.

In the left part of the **Route Planning** panel, press the button to display the corresponding table.



St-Petersburg-Hamburg

WPT	Name	ETA	Stay	Time Zone	ETD	TTG	Total Time	Speed
0	St-Petersburg	XX-XX-XX XX:XX	XX-XX-XX XX:XX	00:00		XX-XX-XX	XX-XX-XX	XXX.X
1				00:00				
2				00:00				
3				00:00				

Edit Options | Anchor | Check Route | Schedule Calculation | Voyage plan | Port Call | Protection control

Advanced Cursors: 90.0, 270.0, 0.10 NM, 99.9 NM

Operations: Reverse, Shift, Link, Delete

Quick Distance Calculator: From WPT: 0 St-Petersburg, To WPT: 86 Hamburg, Distance: 1398.25 NM

To draw a new schedule, switch to the **Schedule calculation** page.

Edit Options | Anchor | Check Route | Schedule Calculation | Voyage plan | Protection control

Schedule calculation

Create Schedule | Clear Schedule | Delete Schedule

Press the **Create Schedule** button in the **Schedule** group.

Schedule

Create Schedule | Clear Schedule | Delete Schedule

Enter schedule elements for WPT's in the table. To do this, perform the procedures detailed below:

- position the cursor on the **ETA** element cell (not set for the route start point). Double click the left trackball button to activate the cell;

ETA

23-09-09
19:18

- enter the ETA for this WPT and press the right trackball button;

ETA

29-09-09
19:00

Apply | Cancel

- press the **Apply** button to confirm the changes you have made;

ETA

29-09-09
19:00

- position the cursor on the **Stay** element cell; double click the left trackball button to activate the cell;

Stay

00 d 00
h 00 m

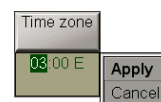
- enter the time of stay at this WPT and press the right trackball button;

Stay

00 d 05
h 00 m

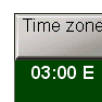
Apply | Cancel

- press the **Apply** button to confirm the changes you have made;
- position the cursor on the Time zone element cell; double click the left trackball button to activate the cell;

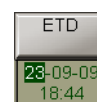


- enter the time zone at this WPT and press the right trackball button;

- press **Apply** button to confirm the changes you have made;

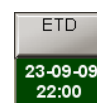


- position the cursor on the **ETD** element cell (not set for the route end point); double click the left trackball button to activate the cell;

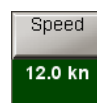


- enter the **ETD** for the WPT and press the right trackball button;

- press the **Apply** button to confirm the changes you have made;



- position the cursor on the **Speed** element cell; double click the left trackball button to activate the cell;

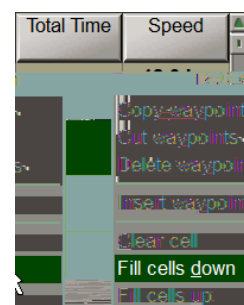


- enter the speed value on the route leg; press the left trackball button.

NOTE: There is no editing of **TTG** and **Total Time** cells.

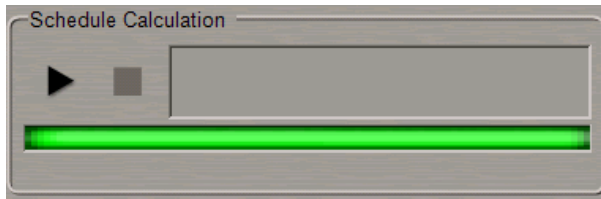
When filling in the table you can use the context menu called by the right trackball button:

- Clear cell** – to delete a value in the selected cell;
- Fill cells down** – to fill in all the cells below with a value entered in the selected cell;
- Fill cells up** – to fill in all the cells above with a value entered in the selected cell.



ATTENTION! For the route start point, the ETD value should always be specified. For the end point, it is necessary to specify ETA or speed of proceeding along the route legs.

Press the  button in the **Schedule Calculation** group: the schedule will be calculated.

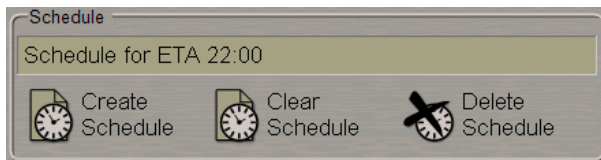


The schedule calculation process is reflected in the display window. In the table, empty cells will be filled in with the calculated schedule elements.

WPT	Name	ETA	Stay	Time Zone	ETD	TTG	Total Time	Speed
0	St-Petersburg	XX-XX-XX XX:XX	XX-XX-XX XX:XX	03:00 E	07-11-17 18:00	XX-XX-XX	XX-XX-XX	XXX.X
1		07-11-17 15:00		00:00	07-11-17 15:00	00 d 00 h 00 min	00 d 00 h 00 min	12.0 kn
2		07-11-17 15:03		00:00	07-11-17 15:03	00 d 00 h 02 min	00 d 00 h 03 min	12.0 kn
3		07-11-17 15:05		00:00	07-11-17 15:05	00 d 00 h 02 min	00 d 00 h 05 min	12.0 kn

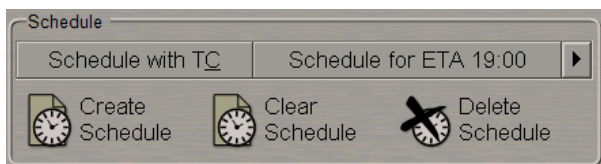
Below the table, the **Schedule Calculation** window shows "Calculation completed" and a green progress bar. The **Schedule** window shows "Schedule 1" and buttons for "Create Schedule", "Clear Schedule", and "Delete Schedule".

A schedule has been created. To save the schedule, enter its name in the name input field of the **Schedule** group.



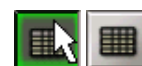
Press <ENTER>.

You can simultaneously create several schedules with different data for one and the same route. As a button with schedule name is pressed, the schedule becomes active and can be edited in the schedule Table. The active schedule can be selected in the line of the **Schedule** group:

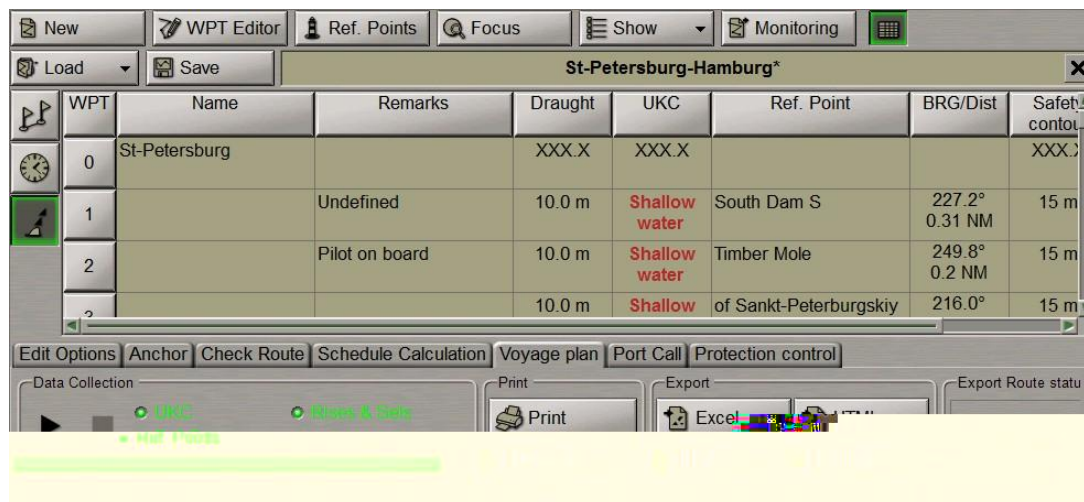


9.3.3. Voyage Plan Data Editing

The display of the table is turned on/off by using the **Show tables** button.



In the left part of the **Route Planning** panel, press the button to display the corresponding table.



The Extra data table contains the following data (columns) on reference points:

- **Ref. point** – reference point name taken from the chart object (if the object has no name, the cell is empty);
- **BRG/Dist** – bearing and distance to the reference point.

Reference points are set automatically during the data collection, if the **Ref. Points** checkbox is selected in the **Data Collection** group of the **Voyage plan** page (see the previous item).

9.3.3.1. Setting and Changing Reference Points

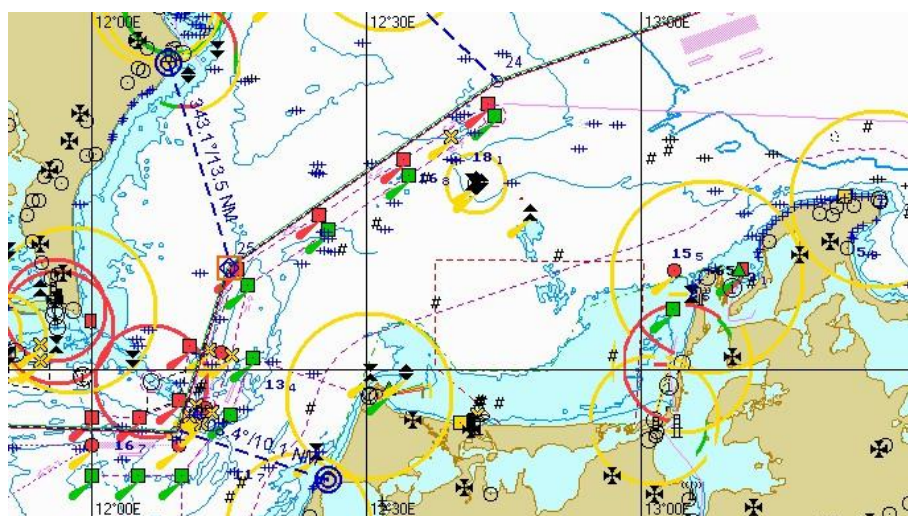
To set or change a reference point, position the cursor on the necessary WPT in the Extra data table and click the left trackball button on the required point.

WPT	Name	Remarks	Draught	UKC	Ref. Point	BRG/Dist	Safety contour
23						12.51 NM	
24			10.0 m	Shallow water	S 78	222.3° 4.8 NM	15 m
25			10.0 m	Shallow water	W 72A	195.2° 5.49 NM	15 m
26			10.0 m	Shallow water	W 70 A	042.5° 0.93 NM	15 m

To set reference points on the Chart Panel, press the **Ref. points** button in the top part of the **Route Planning** panel



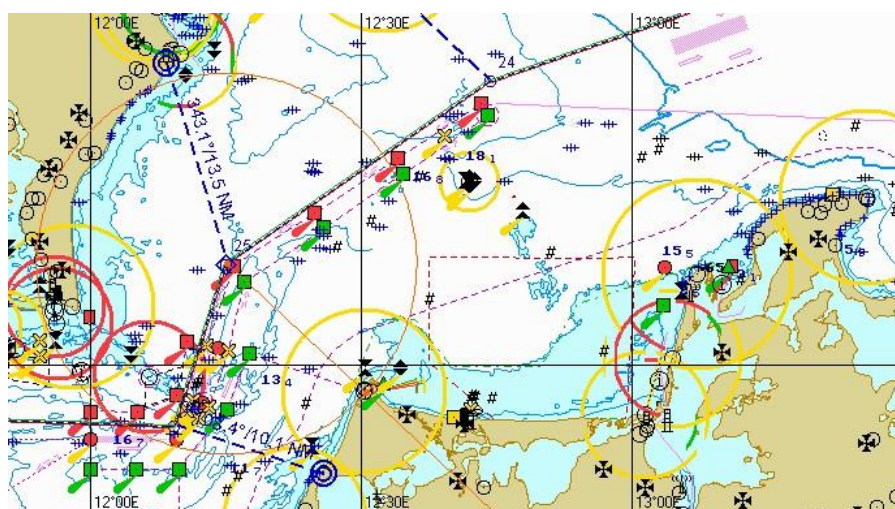
An acquisition marker will appear on the Chart Panel at the selected WPT.



The right-hand part of the screen will display the **Select waypoint** information window showing the acquisition marker coordinates and bearing/distance relative to the WPT WOL.

Select waypoint	
Latitude	54°36.494' N
Longitude	012°14.903' E
Bearing	059.1 °
Opp. Bearing	239.1 °
Range	1134.76 NM
Press Tab to edit manually	

Press the left trackball button: the marker will acquire the WPT and change its own form to the graphics cursor. Position the marker in the new reference point within the displayed Chart Panel. If the **Snap to object** checkbox is selected in the **Route Settings** page of the **Config** panel (see section *Work with Route*, paragraph *Default Route Settings*), the reference point will be referenced to the object automatically (if it is not farther than 0.2 mile from graphics cursor).



The **Select waypoint** information window shows the acquisition marker coordinates and bearing/distance relative to the WPT WOL.

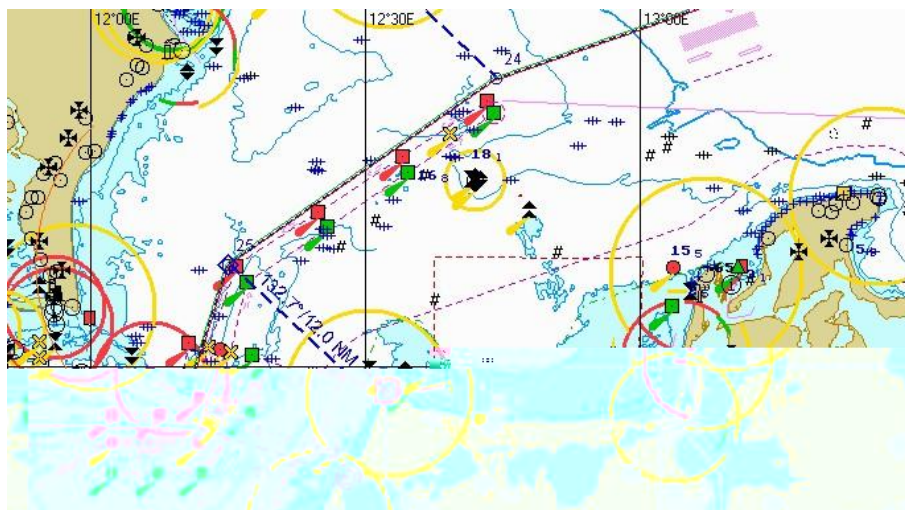
Select waypoint

Latitude 154°36.503' N
Longitude 1012°15.048' E
Bearing134.5 °
Opp. Bearing314.5 °
Range11.86 NM

Latitude 254°28.207' N
Longitude 2012°29.567' E

Press Tab to edit manually

Press the left trackball button.



The reference point will be shown on the Chart Panel in the form of two blue circles. The blue dashed line connects the WPT WOL and its corresponding reference point.

The values of bearing/distance to the reference point from the WPT WOL are shown over the line. The same values are displayed in the table's **BRG/Dist** column.

WPT	Name	Remarks	Draught	UKC	Ref. Point	BRG/Dist	Safety contour
23						12.51 NM	
24			10.0 m	Shallow water	S 78	222.5° 4.83 NM	15 m
25			10.0 m	Shallow water	Darßer Ort	079.9° 12.19 NM	15 m
26			10.0 m	6.2 m		318.3° 7.2 NM	15 m

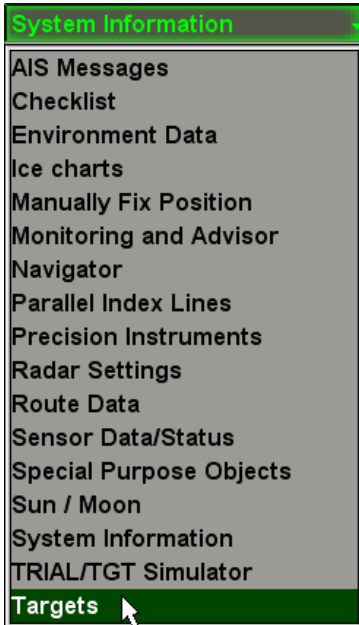
Press the right trackball button to exit from the editing mode. The marker will assume the shape of the free cursor.

10.2.3. ARPA Settings

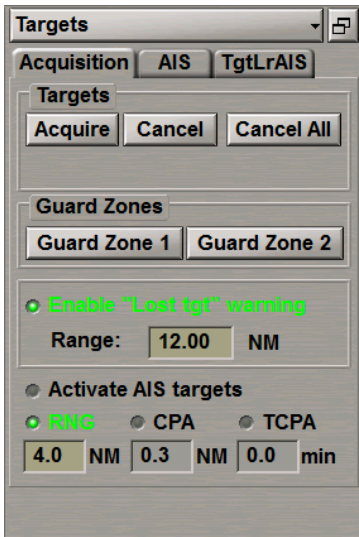
To turn on the display of targets, press the **Overlay** and the **ARPA** buttons in the **Sensors** window of the Control panel.



Press the button with the name of the set display in the **Display Panel** window of the Control panel.



In the list which will open up, select the **Targets** line and press the left trackball button.

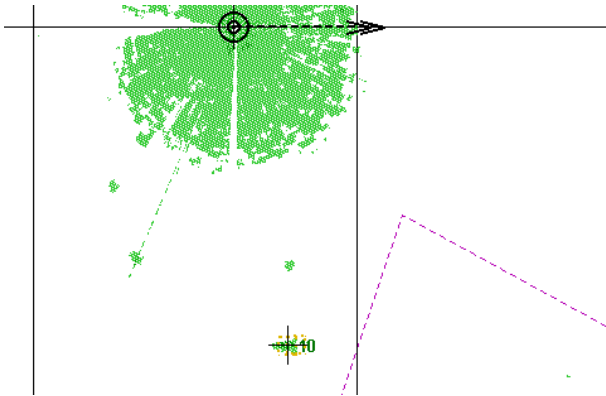


10.2.3.1. To Acquire Targets for Tracking

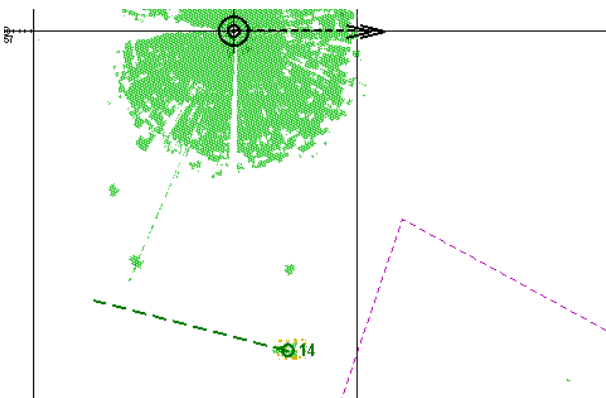
To switch the cursor to the target acquisition mode, press the **Acquire** button in the **Targets** group.



Position the cursor on the target and press the left trackball button.



In a minute, the acquired target will be displayed as a green coloured circle with pre-calculated motion vector.



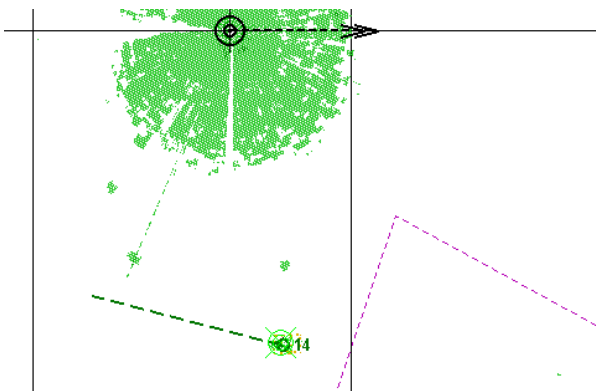
The target has been acquired for tracking. If you press the right trackball button, the cursor exits from the target acquisition mode.

10.2.3.2. To Cancel Tracked Targets

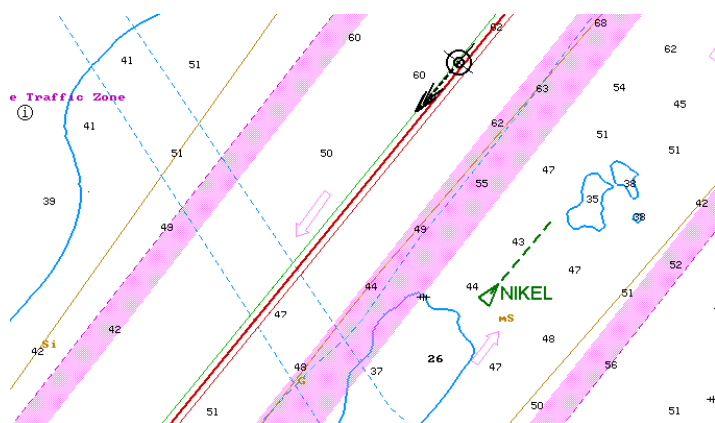
To switch the cursor to the target tracking cancelling mode, press the **Cancel** button in the **Targets** group.



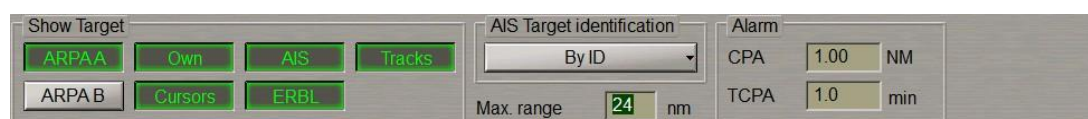
Position the cursor on the tracked target and press the left trackball button.



AIS targets will appear in the Chart panel, and their motion parameters will be displayed in the Targets Table.



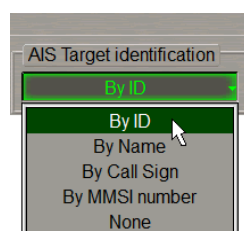
48 nm value is set in the **Max. Range** input box by default, that is the radius of an area within which the AIS targets are processed and displayed.



Set the necessary radius (up to 100 NM) of the AIS targets processing area.

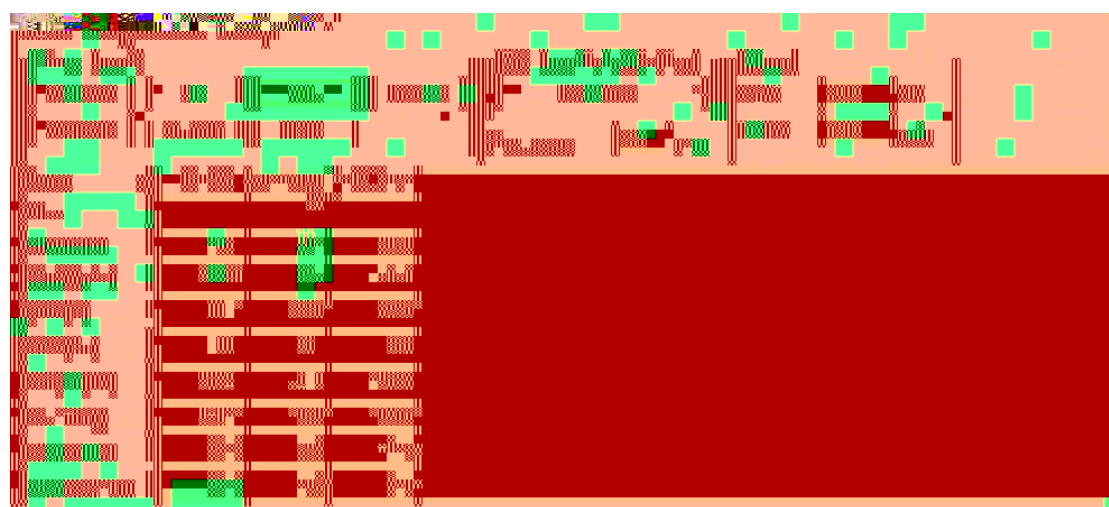
10.3.1.2. Display of AIS Target Identifiers

Press the **AIS Target identification** button to select the AIS targets identifier.



In the list which will open up, select the necessary identifier or **None** to declutter the Chart panel. Press the left trackball button. Targets will be displayed on the Chart panel with the selected identifier, and its name will be shown on the button.

Position the cursor on the target identifier in the Targets Table.

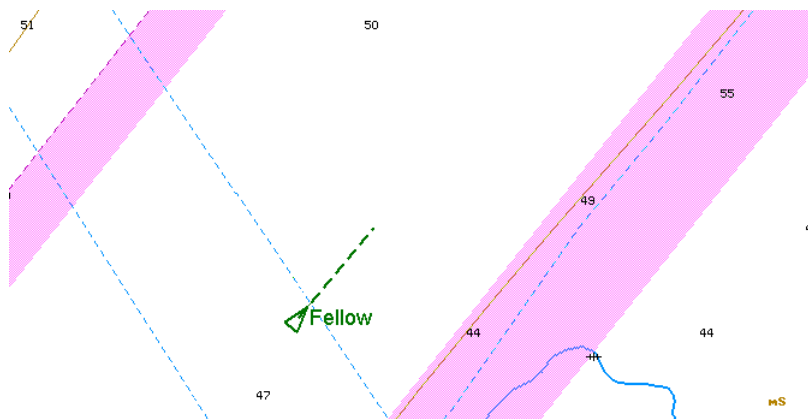


The hint will display the name of the AIS object if it is contained in the messages transmitted by the AIS object.

Enter as required the “Alias” identifier (up to 8 characters) for an AIS object in the **Alias** row of the Targets Table.

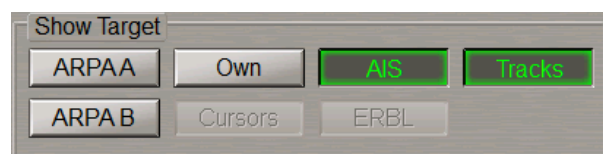
Name	A2	273002414	671641000
Alias		Fellow	
CPA (NM)	1.90	3.24	1.95
TCPA (min)	XXXX	-79.8	XXXX

On the Chart panel, the target will be displayed with the entered “Alias” identifier regardless of settings made in the **AIS Target identification** group. This identifier is displayed on the Chart panel only and is not transmitted in messages.

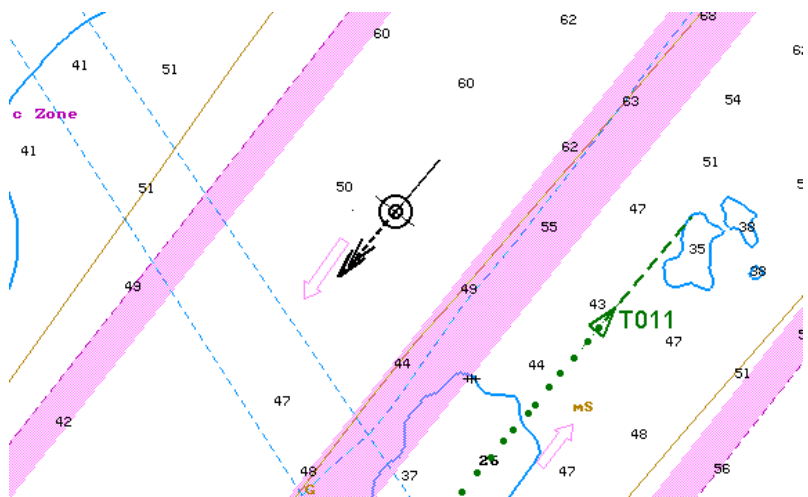


10.3.1.3. Display of Target Tracks

Turn on, as required, the display of target tracks. To do this, press the **Tracks** button in the **Show Target** group.



Target tracks will be displayed on the Chart panel.

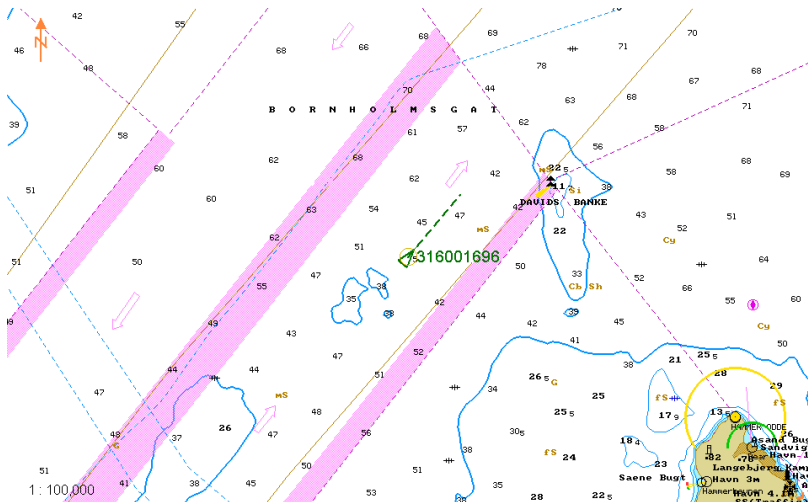


10.3.1.4. Prompt Target Search

Press the button with the target identifier in the **Name** line of the Targets Table, e.g., 316001696.

Name	273002414	316001696	671641000
------	-----------	-----------	-----------

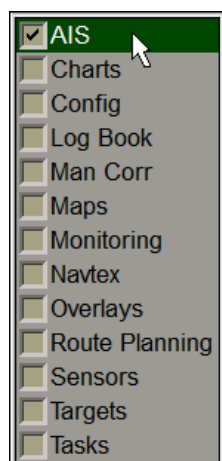
The selected target will be displayed in the centre of the Chart panel highlighted with a flashing circle.



10.3.2. Setting of Voyage Data

ATTENTION! Setting of voyage data is available only on the station with the MASTER status.

Open the **AIS** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Static Data: Restore

Name: KOLA

Call Sign: UCKB

MMSI: 273130500

IMO Number: 008013053

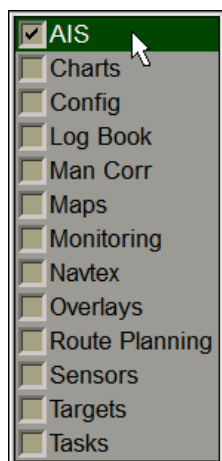
As other devices are connected to the AIS transponder in addition to the MFD (e.g., MKD or pilot software), the own ship name and call sign in the transponder can be changed by these users. The operator is notified about these changes by the **Check AIS Ship Name** and **Check AIS Call Sign** cautions which will remain active until the own ship data is restored in the transponder. To restore the own ship data settings in the transponder as per the settings in the System Configuration utility, press the **Restore** button.

The setting of own ship parameters transmitted in the AIS system is completed.

10.3.3. Transmission of Messages in AIS System

Turn on the display of AIS targets (see paragraph *Handling AIS Targets*).

Open the **AIS** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the left-hand part of the **AIS** panel to switch to the **Messaging** page.

Voyage and Static Data
Messaging
IMO special messages
Interrogation
Long Range
Channel Management

Send message

Create Message

N	Sent to	Date/Time (UTC)	Type	Status
3	316001696	20-04-2006 10:04:49	Safety Text	Waiting
2	273002413	20-04-2006 10:04:27	Safety Text	Delivered
1	ALL	20-04-2006 10:01:43	Safety Text	Sent

Receive message

From: 273002413
To: All stations
Type: Safety text

THE RACON ON LIGHTBUOY TANGO-1
IN PSN 5747.2 N 01046.1 E IS
INOPERATIVE

☒ All message notification
☒ Show new message

Reply

Request Route

Show Sender

⏮

⏪

⏩

⏭

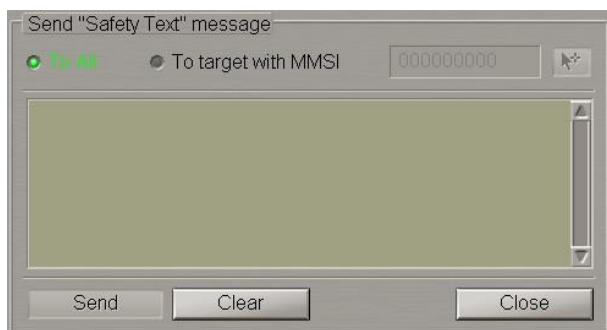
To prepare and transmit a text message, press the **Create Message** button on the **Messaging** page in the **Send message** group and select **Safety Text**.

NOTE: If the class B AIS is selected during the installation, the capability to send messages with the "Safety Text" and/or "Normal Text" status is set in the System Configuration utility. (See *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Installation Guide, Chapter 2, section MFD Configuration, paragraph Sensors Settings*).



Press the left trackball button.

The **Send Message** window will open up, specifying the type of the selected message.



Set the identifier of the addressee of the message to be sent. To transmit a message to all the ships in the radio coverage area, select the **To All** checkbox.



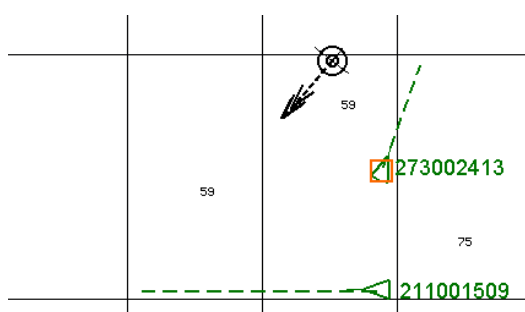
To transmit a message to a certain ship, use one of the three procedures:

1. Select the **To target with MMSI** checkbox and enter the MMSI of the addressee in the input line which will be activated.



2. Select the **To target with MMSI** checkbox and press the  button to the right of the input line.

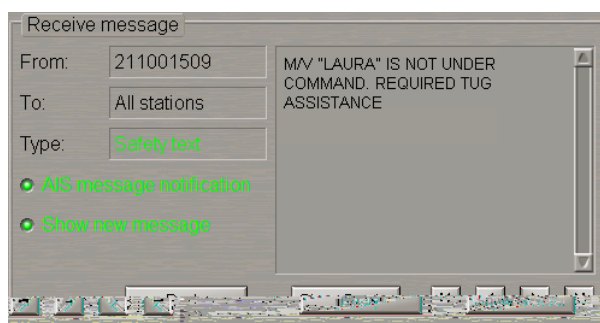
Position the acquisition marker, which will appear on the Chart panel, on the target, which is the addressee of the message.



Press the left trackball button. The input line will display the MMSI of the specified target.



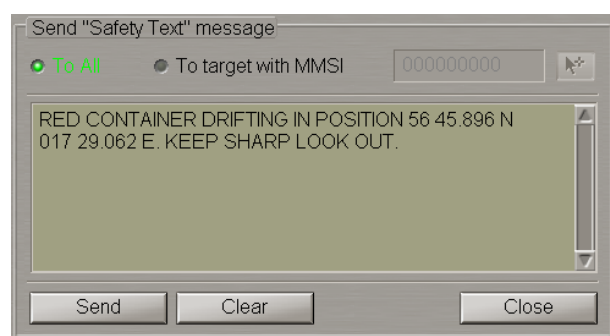
3. To send a reply message, press the **Reply** button in the **Receive Message** group.



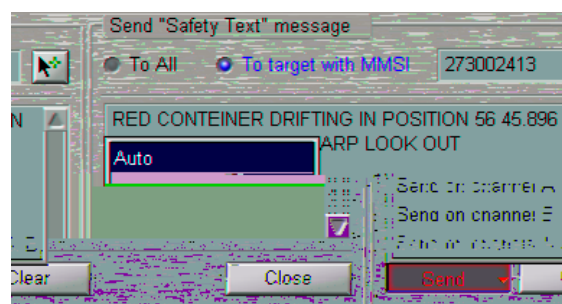
With the use of the **Reply** function, the **Send message** window opens up for a message, which has the same type as the message received from an AIS object, "Safety Text" in this case. The input line will automatically display MMSI of the AIS object, which the message has been received from.



Use the text input window in the **Send Message** group to enter the text of the message which should be sent.



Press the **Send** button.



Select the message despatch channel (**Auto** is generally recommended). The compiled message will be sent.

Press the **Close** button.

Check the sent message status in the table in the **Send message** group. The message status may be one of the following:

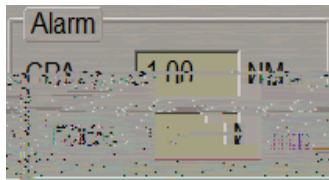
- **Waiting** – the message has been transmitted by the MFD, but no confirmation of its relay has been received from the transponder;
- **Sent** – the message has been sent by the transponder, but no confirmation of its delivery to the addressee has been received or confirmation is not expected if the message was sent to all the AIS objects (“All” identifier);
- **Delivered** – the message has been delivered to the addressee;
- **Failed** – the message, for some reasons, cannot be sent by the transponder.

The text of sent “Safety Text” messages, as well as rected

Use the tab in the top part of the **Targets** panel which will open up, to switch to the **Target Table** page:

Name	40	41	42	43	44	37	47	48	49	50	51	52	53	54
Alias														
CPA (NM)	2.88	5.87	7.32	6.81	7.74	5.81	4.09	4.53	5.86	8.23	1.25	6.76	8.07	5.62
TCPA (min)	-9.7	-8.4	-6.9	-14.5	-9.3	-21.2	-21.2	-21.2	10.4	12.8	21.8	-16.4	-16.6	-19.2
SOG (kn)	45.3	43.5	29.4	26.8	33.1	11.3	15.0	13.1	22.4	12.3	16.8	19.5	15.8	12.8
COG (°)	14.5°	295.9°	65.7°	287.1°	281.1°	30.0°	5.9°	36.4°	84.2°	170.7°	81.6°	348.7°	358.2°	29.6°
Range (NM)	8.94	8.33	8.43	9.10	9.05	8.79	8.67	8.57	7.63	8.46	8.44	9.59	10.11	8.56
Bearing (°)	359.3°	350.7°	0.3°	351.7°	352.5°	352.7°	45.2°	5.8°	203.4°	236.6°	257.5°	317.5°	318.4°	352.4°
Bow X (NM)	7.7	-5.9	-20.9	-6.9	-8.1	55.4	-10.5	112.1	10.6	8.2	-2.6	11.1	16.8	49.3
TBow X (min)	-17.9	-9.0	25.4	-16.5	-13.8	-197.9	5.6	-346.8	-8.4	15.6	27.7	-37.6	-56.9	-164.8

In the top of the **Targets** panel which will open up, turn on the display of targets from the required sources by pressing the appropriate buttons: **ARPA A(B)**, **Own** or **AIS**.



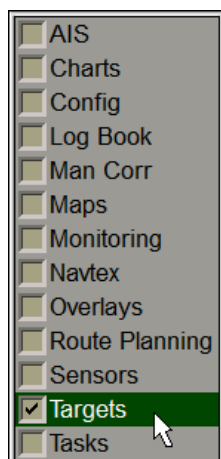
The **Alarm** group contains windows for entering CPA/TCPA values.

The **CPA/TCPA Tgt ID** alarm generation is turned on when both CPA and TCPA values of the target are smaller than the set values:

- **CPA** – to enter the minimum closest point of approach distance;
- **TCPA** – to enter the minimum time to the closest point of approach.

10.4.2. Handling Target Table

Open the **Targets** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Targets** panel which will open up, to switch to the **Target Table** page:

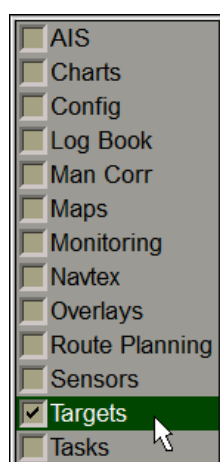
Name	71	74	75	77	78	79	80	91	92	93	95	100	138	182
Alias	Fellow													
CPA (nm)	3.08	2.39	2.01	1.04	0.74	0.07	1.94	3.29	3.70	3.67	1.49	1.33	3.45	0.64
TCPA (min)	-2.5	11.5	1.1	-3.1	-2.3	1.3	3.7	0.8	0.5	-0.3	-1.9	1.4	1.3	8.9
COG (°)	223.6°	335.2°	354.1°	202.5°	219.0°	340.8°	222.2°	268.9°	264.1°	244.8°	292.1°	23.1°	336.6°	336.7
SOG (kn)	91.7	8.5	32.6	13.6	9.5	9.3	5.3	37.1	42.2	51.0	39.5	22.9	37.5	19.1
Range (nm)	5.18	3.82	2.11	1.46	1.00	0.34	2.12	3.35	3.72	3.68	2.16	1.41	3.57	3.75
Bearing (°)	264.4°	158.2°	85.4°	185.3°	197.6°	135.9°	140.1°	10.3°	2.0°	334.9°	331.3°	107.4°	68.9°	144.8
Bow X (nm)	-4.6	-4.8	2.2	1.6	1.8	-0.1	7.0	-213.1	-44.4	-10.2	4.9	1.3	4.3	-0.9
TBow X (min)	-0.5	27.6	-0.3	-7.0	-7.7	1.7	-25.0	272.3	51.4	9.2	-7.7	1.2	-2.2	10.5

The Target Table contains columns (ARPA and AIS targets) and rows with the targets information with a capability to sort by the relevant parameter:

- **Name** – target identifier;
- **Alias** – target identifier (up to 8 characters) assigned by the operator (the target is displayed on the Chart panel with this identifier regardless of settings in the **AIS Target Identification** group);
- **CPA** – closest point of approach distance;
- **TCPA** – time to the closest point of approach;
- **COG** – target course over the ground;
- **SOG** – target speed over the ground;
- **Range** – distance to the target;
- **Bearing** – bearing to the target;
- **Bow X** – distance to the point where the target crosses the own ship course;
- **TBow X** – time to the point where the target crosses the own ship course.

10.4.3. Prompt Target Search

Open the **Targets** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



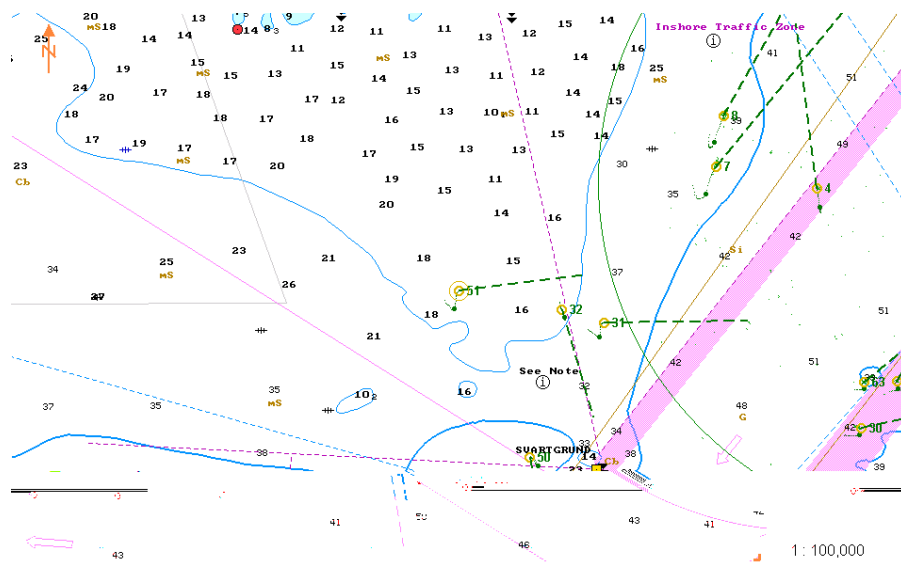
Use the tab in the top part of the **Targets** panel which will open up, to switch to the **Target Table** page:

Target Table		AIS color tracks	
Show Target		AIS Target identification	
ARPA A	ARPA B	By Name	Alarm
ARPA B	Cursors	Max. range	CPA 2.00 NM
	ERBL	48 nm	TCPA 12.0 min
Name	40	41	42
Alias			
CPA (NM)	2.88	5.87	7.32
TCPA (min)	-9.7	-8.4	-6.9
COG (°)	14.5°	295.9°	65.7°
SOG (kn)	45.3	43.5	29.4
Range (NM)	8.94	8.33	8.43
Bearing (°)	359.3°	350.7°	0.3°
Bow X (NM)	7.7	-5.9	-20.9
TBow X (min)	-17.9	-9.0	25.4

Turn on the display of targets (see section *AIS*, paragraph *Handling AIS Targets* and section *ARPA*, paragraph *Target Display*).

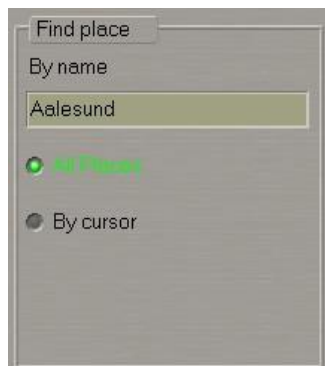
Press the button with the target identifier in the **Name** line of the Targets Table, e.g., “51”.

The selected target will be displayed in the centre of the Chart panel highlighted with a flashing circle.



In the right-hand part of the **Place** tab which will open up, select the reference point by using one of the following procedures:

- Enter the reference point name in the **By name** input line and press <ENTER>;
- Select the **All Places** checkbox.

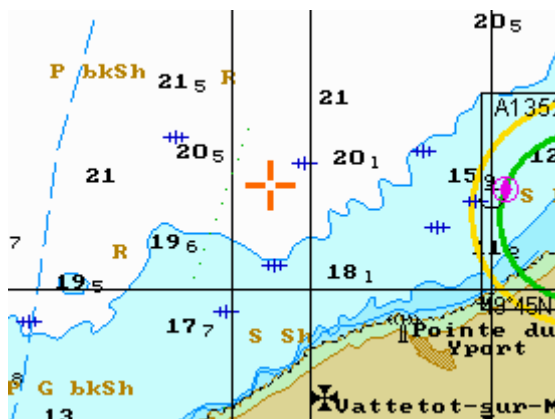


In the list in the left-hand part of the page which will open up, select the reference point and double click the left trackball button.

- Select the **By cursor** checkbox; enter the radius of the area coverage circle in the **Distance** input line which will open up.



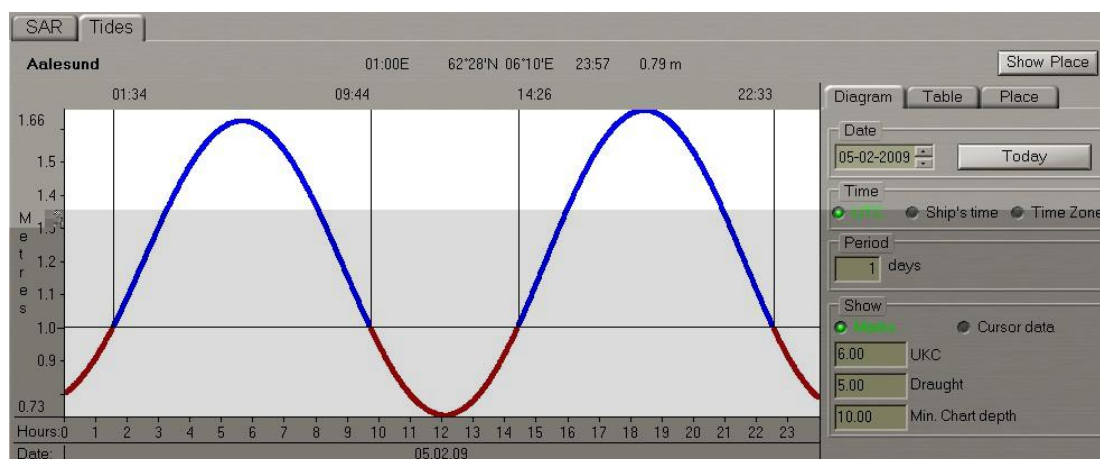
Press the **Find** button. Position the graphic cursors which will appear, on the point within the Chart panel which will be the centre of the area coverage circle.



Press the left trackball button. In the list of reference points which fall within the coverage area in the left-hand part of the page, select the reference point and double click the left trackball button.

11.5.1.1. Information Displayed Graphically

After the selection of the reference point, the **Diagram** tab will open up, displaying the tidal curve for the selected reference point.



In the **Date** line of the **Diagram** tab, select the date starting from which the tidal heights will be calculated. Use the **Period** input line to enter the period which tidal heights are required to be calculated for.

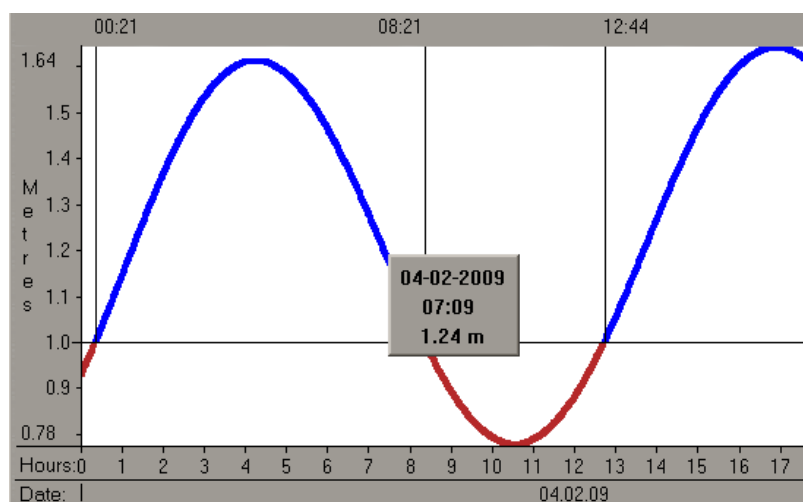
The following parameters can be set:

- **UKC** – under keel clearance;
- **Draught** – ship's draught;
- **Min. Chart depth** – the minimum depth on the chart.

Select the **Marks** checkbox to show the time marks on the diagram.

On the graph, blue is used for showing the time periods when the actual depth is safe, and red for showing the time intervals with dangerous depths.

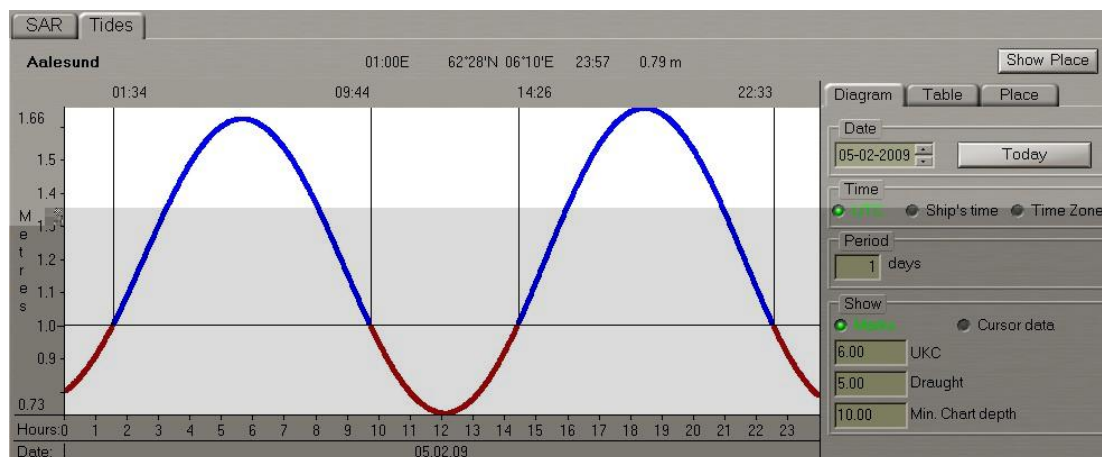
Select the **Cursor Data** checkbox. Position the cursor on the tidal curve and by moving the cursor with the trackball, set the necessary time value in the cursor information window.



Read the tidal height value in the cursor information window.

11.5.1.2. Information Displayed in Tabular Form

After the selection of the reference point, the **Diagram** tab will open up displaying the tidal curve for the selected reference point.



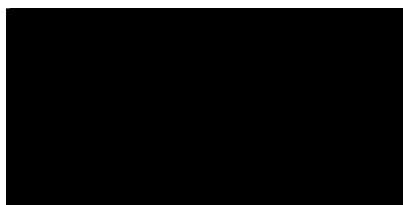
Use the tab in the top right part of the page to switch to the **Table** tab.

Heights of water in Metres for high and low waters	
Aalesund	
62°28'N 06°10'E	
UTC TIME	
4 February 2009	
04:15	1.62
10:33	0.78
16:55	1.64
23:23	0.78

In the **Date** line of the **Table** tab, select the date starting from which the tidal heights will be calculated. Use the **Period** input line to enter the period which tidal heights are required to be calculated for.

To calculate tidal tables, use one of the following procedures:

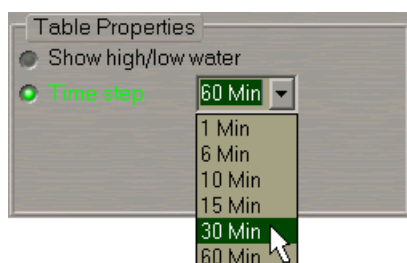
- In the Table Properties group, select the Show high/low water checkbox.



The left-hand part of the page will display a table of high and low water.

Heights of water in Metres for high and low waters	
Aalesund	
62°28'N 06°10'E	
UTC TIME	
4 February 2009	
04:15	1.62
10:33	0.78
16:55	1.64
23:23	0.78

- In the **Table Properties** group, select the **Time step** checkbox; press the  button in the activated line for the input of the tidal height calculations discretion.



In the list which will open up, select the necessary tidal height calculation discretion and press the left trackball button. The left-hand part of the page will display the tidal height table calculated with the set discretion.

Heights of water in Metres for regular intervals	
Aalesund	
62°28'N 06°10'E	
UTC TIME	
4 February 2009	
00:00 0.93	04:00 1.61
00:30 1.03	04:30 1.61
01:00 1.15	05:00 1.59
01:30 1.26	05:30 1.54
02:00 1.37	06:00 1.46
02:30 1.46	06:30 1.38
03:00 1.54	07:00 1.28
03:30 1.59	07:30 1.17
08:00 1.07	11:00 0.79
08:30 0.97	11:30 0.82
09:00 0.89	12:00 0.88
09:30 0.83	12:30 0.96
10:00 0.79	13:00 1.05
10:30 0.78	13:30 1.16
11:00 0.79	14:00 1.27
11:30 0.82	14:30 1.37
12:00 0.88	15:00 1.47
12:30 0.96	15:30 1.55
13:00 1.05	16:00 1.60
13:30 1.16	16:30 1.64
14:00 1.27	17:00 1.64
14:30 1.37	17:30 1.63
15:00 1.47	18:00 1.59
15:30 1.55	18:30 1.52
16:00 1.60	19:00 1.44
16:30 1.64	19:30 1.34
17:00 1.64	20:00 1.24
17:30 1.63	20:30 1.13
18:00 1.59	21:00 1.03
18:30 1.52	21:30 0.94
19:00 1.44	22:00 0.87
19:30 1.34	22:30 0.81
20:00 1.24	23:00 0.78
20:30 1.13	23:30 0.78

Take the necessary readings in the calculated tables.